A discrete $\leftrightarrow$ continuous bridge for the Riemann zeta and Dirichlet eta functions: a zero-birth knob, a midpoint warp onto $\zeta(s, \frac{1}{2})$, and a unified $O(1/K)$ rate law

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Abstract

We study a one-parameter family of analytic objects that interpolates from the bland continuous integral $\int_1^\infty x^{-s}\,\mathrm{d}x = 1/(s-1)$ (which keeps ζ's pole but has *no* non-trivial zeros, no functional equation, and no arithmetic) to the full Dirichlet series $\zeta(s)$. The interpolation knob is the number K of Fourier harmonics of the discreteness kernel (the Euler–Maclaurin sawtooth, equivalently the Abel–Plana comb) that one retains. As K grows, the non-trivial zeros are *born* out of a literally zeroless endpoint and then *migrate* onto their critical lines. We exhibit two routes to the same destination, an *additive* comb that adds harmonics to the integrand and a *nonlinear midpoint warp* of the integration variable that is deliberately engineered to land on the half-shifted Hurwitz zeta $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$; and we record the general-phase warp onto $\zeta(s, \alpha)$, which reproduces the Saias–Weingartner $P \cdot L$ dichotomy (clean vertical zero strings only at $\alpha \in \{0, \frac{1}{2}, 1\}$). The unifying result is a single $O(1/K)$ *rate / birth-K law*: every route truncates the same sawtooth, so every route converges at the same algebraic rate, with a route-dependent displacement constant whose real part fixes the approach side and whose imaginary part is numerically tied to Gram's law. The same law, followed onto the negative real axis, also accounts for the *trivial* zeros: they surface as a receding $O(1/K)$ tide uncovers $-2, -4, -6, \ldots$, the first inherited from the endpoint and the rest born as conjugate pairs that pinch onto the axis. Finally we read the Dirichlet-η zeros as a two-component "crystal $\times$ GUE" spectrum and isolate the single prime $p = 2$. The construction belongs to a half-century-old lineage of deforming a Dirichlet series and following where its zeros go; its point of departure is that the harmonic-count knob makes those zeros *converge* onto the critical line and stay there, rather than merely graze it in passing. This is a *computational / experimental-mathematics* synthesis: every individual endpoint is classical and is cited as such; the contribution is the unifying construction, the rate law, and the spectral readings, not a new theorem about where zeros live. An appendix carries the picture to four families beyond ζ and η, reproducing the construction's published deform-and-track ancestors and running its own knob forward on a primitive Dirichlet L-function. All figures and numbers are reproducible from the accompanying repository.

Contents

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1 Introduction

The Riemann zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ is a *discrete* object: a sum over the integers. Its simplest continuous shadow is the integral that replaces the sum by its leading Euler–Maclaurin term,

$$\int_1^\infty x^{-s}\,\mathrm{d}x = \frac{1}{s-1}, \qquad \operatorname{Re} s > 1, \tag{1}$$

which retains ζ's pole at $s = 1$ but is otherwise inert: it has *no* non-trivial zeros, no functional equation, and no arithmetic content whatsoever. Replacing a discrete arithmetic sum by such a continuous integral is the general "continuous model" of an arithmetic function [34]; (1) is its ζ instance, and the zeroless starting point of everything that follows. The gap between (1) and the genuine $\zeta(s)$ is the discreteness of the integers, and the Euler–Maclaurin formula makes that gap explicit as a sum of correction terms built from the periodic Bernoulli “sawtooth” $\widetilde{B}_1(x) = \{x\} - \frac{1}{2}$. The Fourier series of the sawtooth,

$$\widetilde{B}_1(x) = \{x\} - \tfrac{1}{2} = -\sum_{k\geq 1} \frac{\sin(2\pi k x)}{\pi k}, \tag{2}$$

suggests an obvious interpolation knob: keep only the first K harmonics. At $K = 0$ one is back at the bland endpoint (1); as $K \to \infty$ one restores the full discreteness and recovers $\zeta(s)$.

This paper studies what the non-trivial zeros do along that interpolation. The short answer, made literal and visual throughout, is that the zeros are *born* as K leaves 0 and then *migrate* onto their critical lines as $K \to \infty$. Because the $K = 0$ endpoint is genuinely zeroless, this is a *creation-and-migration* mechanism, not a rearrangement of a pre-existing zero set; we contrast it carefully with the de Bruijn–Newman heat flow in Section 10.

The idea of deforming a Dirichlet series by a parameter and tracking its zeros is itself classical, and its direct ancestor is the study of the complex zeros of the polylogarithm by Fornberg and Kölbig [10], which sweeps the polylogarithm’s argument from -1 to $+1$ (the endpoints being $-\eta$ and ζ) and numerically follows the zero trajectories. There the moving zeros only *graze* the critical line before being absorbed into the pole that emerges at the endpoint; the harmonic-count knob studied here instead drives them to *converge* onto it and stay. We reproduce that contrast, the closest published deform-and-track analogues, and a reverse-direction foil, and run the bridge’s own knob forward on a primitive Dirichlet L-function—four families beyond ζ and η—in Appendix A.

Honest framing. We state at the outset, as a deliberate decision, that this is a *methods / experimental-mathematics* paper: a unified, reproducible re-derivation that threads several *known* endpoints onto one explicit “restore discreteness” knob and measures how the zeros respond. It is *not* a theorem about where the zeros of ζ live, and it carries no implication toward the Riemann hypothesis. Every individual ingredient below is classical, and we have made two adversarial passes through the literature to keep the attribution honest. What is classical, and what we believe is the surviving synthesis, is stated plainly here and revisited in Section 10:

- **Classical (we cite, we do not claim):** the half-shift identity $(2^s - 1)\zeta(s) = \zeta(s, \frac{1}{2})$ and the migration of Hurwitz-type zeros onto $\sigma = \frac{1}{2}$ [11, 16]; the privileged $\alpha \in \{0, \frac{1}{2}, 1\}$ $P \cdot L$ dichotomy [30], with generic off-line $\zeta(s, \alpha)$ zeros [6, 8]; the $\sigma = 1$ eta comb at $t = 2\pi k/\ln 2$ [2, 32]; and the appearance of prime-power log-frequencies (smallest $\ln 2$) in the zeta-zero number variance, in a regime where the GUE prediction fails [3, 20].
- **The synthesis we offer:** the harmonic-count K as a *zero-birth knob* attached to a literally zeroless endpoint; the Euler–Maclaurin sawtooth $\equiv$ Abel–Plana comb identity read as a zero-birth event; the midpoint warp *deliberately* landing on $\zeta(s, \frac{1}{2})$ (with its $\sigma = 0$ companion as the functional-equation mirror of η’s $\sigma = 1$ comb); a *unified $O(1/K)$ rate law* tying the additive and warp routes together, with a Gram’s-law coupling on the vertical axis; and a comb $\uplus$ GUE two-component reading of the η zeros that isolates the single prime $p = 2$.

Roadmap. Section 2 sets up the two endpoints: the bland $1/(s-1)$ and the structured continuous-η integral, whose off-critical zero string already obeys the Riemann–von Mangoldt counting law. Section 3 then places the two restoration routes side by side at the level of the integrand—an added kernel versus a warped variable—before either is developed. Section 4 restores discreteness *additively* and exhibits the zero-migration map. Section 5 restores it by a nonlinear *warp* of the integration variable, deliberately targeting $\zeta(s, \frac{1}{2})$, and leads with the algebra that makes the half-integer object *contain* ζ; a short supplement (Section 5.5) warps the η integrand itself and finds the midpoint phase annihilated by its cos sign-carrier. Section 6 generalizes the warp to arbitrary phase and recovers the $P \cdot L$ dichotomy. Section 7 is the unifying $O(1/K)$ rate law, with a stability corollary (Section 7.1): every finite-K interpolant is meromorphic on all of $\mathbb{C}$, with no abscissa of convergence. A short section (Section 8) then follows the negative real axis and watches the *trivial* zeros be born there. Section 9 reads the η zeros as a two-component spectrum. Section 10 foregrounds the prior-art deflators and gives the de Bruijn–Newman contrast, and Section 11 points to the reproducible code. Appendix A, a self-contained appendix, carries the picture to four families beyond ζ and η: reproducing the published deform-and-track ancestors and a reverse-direction foil, and running the bridge's own knob on a primitive Dirichlet L-function. A short closing appendix (Appendix B) runs the construction on the geometric series, where it reduces entirely to elementary closed form—a consistency check, not a result.

2 The two endpoints

2.1 The bland endpoint

For $\operatorname{Re} s > 1$, (1) is the entire continuous content of the leading Euler–Maclaurin term applied to ζ. Analytically continued, $1/(s-1)$ is a meromorphic function with a single simple pole and *no zeros at all*. It is the honest "$K = 0$" object on the ζ side: pole present, everything else absent.

2.2 The structured continuous endpoint

The alternating (Dirichlet) eta function $\eta(s) = \sum_{n\geq 1}(-1)^{n-1}n^{-s} = (1-2^{1-s})\zeta(s)$ has a more interesting continuous shadow, because the alternating sign is $(-1)^n = \cos(\pi n)$ at the integers. Replacing the sum by the corresponding integral gives

$$F(s) \;=\; \int_1^\infty \cos(\pi x)\, x^{-s}\, \mathrm{d}x, \tag{3}$$

which, unlike (1), is genuinely structured: it has a closed form in incomplete gamma functions and a non-trivial zero string. Numerically (see Fig. 1) the zeros of (3) obey three laws, confirmed to 3–5 digits over $8 < t < 160$:

$$\text{(real part)} \quad \sigma(t) \to \tfrac{3}{2}, \tag{4}$$

$$\text{(spacing)} \quad \Delta t \sim \frac{2\pi}{\ln(t/\pi)}, \tag{5}$$

$$\text{(counting)} \quad N(t) = \frac{t}{2\pi}\ln\frac{t}{\pi \mathrm{e}} + \frac{5}{8}, \tag{6}$$

the last of which is the Riemann–von Mangoldt counting form familiar from ζ itself [27, 35].

Observation 1. The continuous endpoint already supplies non-trivial zeros with the *correct Riemann–von Mangoldt density* (6); what it withholds is their *location*. The zeros live off the critical line, drifting toward $\sigma = \frac{3}{2}$. "Zeros with the right counting law" is therefore a property of the continuum; "those zeros on $\operatorname{Re} s = \frac{1}{2}$" is what restoring discreteness must supply. Closing that gap is the whole point of the bridge.

3 Two routes, side by side: an added kernel vs. a warped variable

The next two sections restore discreteness by two different routes, and the distinction between them is the single most useful thing to see clearly before either is developed, so we draw it first.

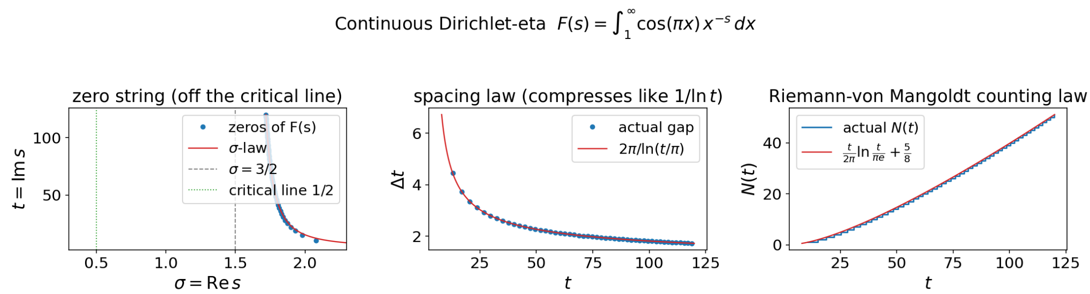

Figure 1: The continuous-η endpoint $F(s) = \int_1^\infty \cos(\pi x)x^{-s}\mathrm{d}x$ and its off-critical zero string. The zeros have the *right density* (the Riemann–von Mangoldt law (6)), but their real parts drift toward $\sigma = \frac{3}{2}$ (4): the continuum supplies "non-trivial zeros with the correct counting law" but not "zeros on the critical line."

Both routes inject the *same* K-harmonic discreteness kernel,

$$\phi_K(x) \;=\; \sum_{k=1}^{K} \frac{\sin(2\pi k x)}{\pi k}, \tag{7}$$

the Fourier partial sum of the sawtooth $\frac{1}{2} - \{x\}$ (so $\phi_K \to \frac{1}{2} - \{x\}$ as $K \to \infty$, and $\phi_0 \equiv 0$). What differs is only *where* ϕ_K enters the integrand of the ζ-side object.

The **additive comb** (Section 4) leaves the integration variable alone and adds the kernel to the integrand. Euler–Maclaurin on ζ with $g(x) = x^{-s}$ gives, term by term,

$$\zeta^{(K)}(s) \;=\; \int_1^\infty \Big[x^{-s} + s\,\phi_K(x)\,x^{-s-1}\Big]\mathrm{d}x \;+\; \tfrac{1}{2}, \tag{8}$$

since $\sum_{k\le K} \frac{s}{\pi k} \int_1^\infty \sin(2\pi k x)\, x^{-s-1}\mathrm{d}x \;=\; \int_1^\infty s\,\phi_K(x)\,x^{-s-1}\mathrm{d}x$ (the $+\frac{1}{2}$ is the constant Euler–Maclaurin endpoint term, a boundary contribution, not part of the integrand). The integrand $x^{-s} + s\,\phi_K(x)\,x^{-s-1} = x^{-s}(1 + s\,\phi_K(x)/x)$ is the smooth x^{-s} carrying an *added* K-harmonic ripple; it is *linear* in ϕ_K, and the variable x is untouched.

The **variable warp** (Section 5) instead leaves the integrand's shape alone and composes the kernel into the argument, $x \mapsto x + \phi_K(x)$:

$$W_K(s) \;=\; \int_1^\infty \big(x + \phi_K(x)\big)^{-s}\mathrm{d}x. \tag{9}$$

This is *nonlinear* in ϕ_K. As $K \to \infty$, $x + \phi_K(x) \to \lfloor x \rfloor + \frac{1}{2}$, so each unit cell collapses onto its midpoint and the integrand becomes the midpoint staircase $(\lfloor x \rfloor + \frac{1}{2})^{-s}$, which is why the warp lands on the half-shifted Hurwitz zeta $\zeta(s, \frac{1}{2})$ (Section 5).

The contrast is then a one-line picture (Fig. 2):

$$\begin{array}{c|c} \textbf{additive comb} & \textbf{variable warp} \\ x^{-s} + s\,\phi_K(x)\,x^{-s-1} & \big(x + \phi_K(x)\big)^{-s} \\ \text{kernel } \textit{added}\text{; } x \text{ linear} & \text{kernel } \textit{composed in}\text{; } x \text{ warped} \end{array} \tag{10}$$

Both reduce to the same bland endpoint at $K = 0$ (then $\phi_0 \equiv 0$, both integrands are x^{-s}, and both objects are $1/(s-1)$, the comb's with its $+\frac{1}{2}$), and both restore full discreteness as $K \to \infty$, but by *addition* versus *composition*. That shared truncated kernel (7) is also why the two routes converge at the *same* $O(1/K)$ rate (Section 7), differing only in the displacement constant. A companion view of the warp *coordinate* $x + \phi_K(x)$ itself, bending from the straight line into the midpoint staircase, appears in Section 5 (Fig. 5).

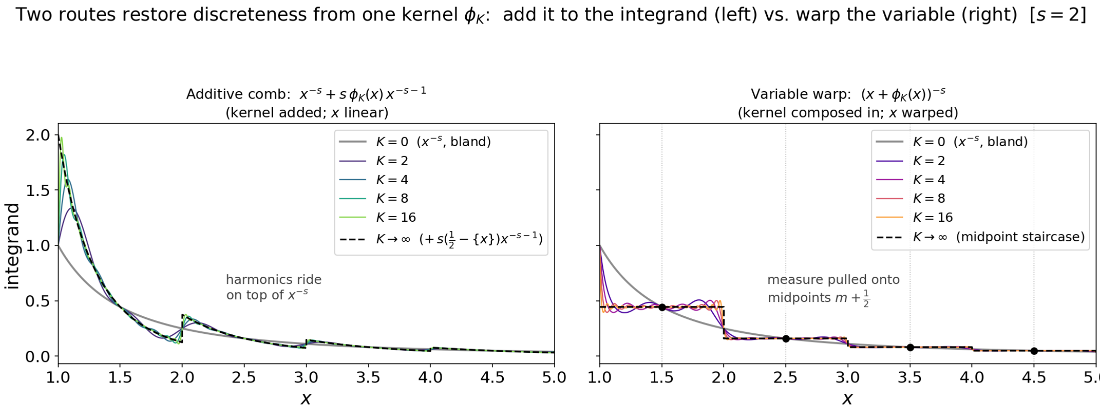

Figure 2: The two restoration routes drawn at the level of the integrand, at a real $s = 2$ and harmonic counts $K = 2, 4, 8, 16$. *Left:* the additive comb $x^{-s} + s\,\phi_K(x)\,x^{-s-1}$ (8) – the same kernel ϕ_K is *added* on top of the smooth x^{-s} (grey, $K = 0$), an oscillation that converges to the full sawtooth ripple (dashed). *Right:* the variable warp $(x + \phi_K(x))^{-s}$ (9) – the same kernel is *composed into the argument*, pulling the measure onto the cell midpoints $m + \frac{1}{2}$ (dots) so the integrand collapses toward the midpoint staircase (dashed). Same kernel, same K, same $K = 0$ start; addition on the left, composition on the right.

4 Phase 1: the additive comb

The first route restores discreteness *additively*: add the Fourier harmonics (2) of the discreteness kernel back into the integrand, one at a time, and track each zero as a curve in K. We call the resulting trajectories the *zero-migration map* (Fig. 3).

Two integral guises of one comb. The discreteness correction admits two classical integral representations that we verify agree term by term: the Euler–Maclaurin sawtooth $\widetilde{B}_1(x) = \{x\} - \frac{1}{2}$ and the Abel–Plana "Bose" comb $1/(\mathrm{e}^{2\pi x} - 1)$. They are the same object viewed through two windows; truncating either at K harmonics gives the identical interpolant. We read this shared truncation as the engine of the rate law in Section 7.

Birth on the ζ side. On the ζ side the zeroless endpoint is $1/(s-1) + \frac{1}{2}$ (the integral plus the constant Euler–Maclaurin term). As the *first* harmonic switches on, the non-trivial zeros are literally *born*, and they then slide onto $\sigma = \frac{1}{2}$ as K increases. There is no zero to "move" at $K = 0$; the zero set is created.

Splitting on the η side. On the η side the $K = 0$ ground string *splits* as K grows: half the zeros onto $\sigma = \frac{1}{2}$ (the genuine ζ zeros) and half onto $\sigma = 1$ (the zeros of the prefactor $1 - 2^{1-s}$, located at $t = 2\pi m/\ln 2$). This split is a near-bijection of densities: writing the two pieces out,

$$\underbrace{\frac{t}{2\pi}\ln\frac{t}{2\pi\mathrm{e}}}_{\sigma=1/2\ \text{(critical line)}} + \underbrace{\frac{t\ln 2}{2\pi}}_{\sigma=1\ \text{(prefactor comb)}} = \frac{t}{2\pi}\ln\frac{t}{\pi\mathrm{e}}, \tag{11}$$

which recovers the continuous endpoint's density (6). No zeros are created or destroyed in the η migration; the existing string is *sorted* onto two lines, and the driver's counting check witnesses (11) numerically.

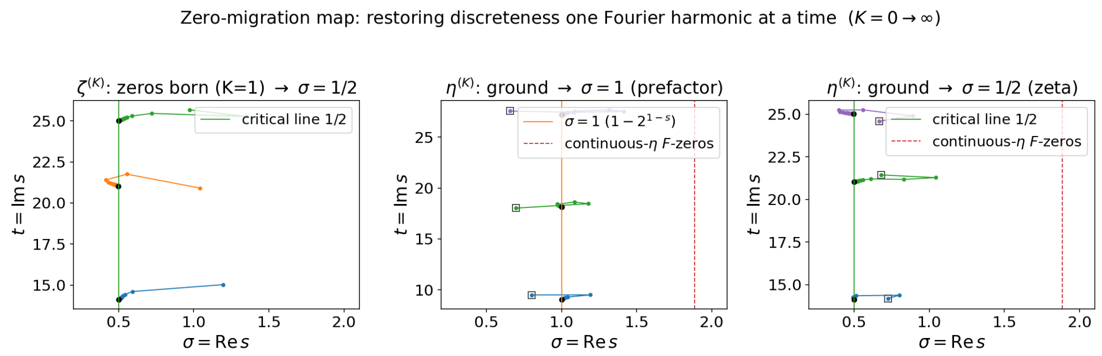

Figure 3: The additive zero-migration map: restoring discreteness one harmonic at a time. On the ζ side the zeros are *born* as the first harmonic switches on and slide onto $\sigma = \frac{1}{2}$; on the η side the ground string *splits* onto $\sigma = \frac{1}{2}$ (the ζ zeros) and $\sigma = 1$ (the $1 - 2^{1-s}$ comb at $t = 2\pi m/\ln 2$), conserving the density (11).

5 Phase 2: the nonlinear midpoint warp

The second route reaches the critical line by a different mechanism: instead of adding harmonics to the integrand, we *warp the integration variable* so the measure is pulled toward integer plateaus, $n^* = x + \varphi_K(x)$, where φ_K is the K-harmonic truncation of the sawtooth displacement. As $K \to \infty$ the warp collapses each unit cell onto its *midpoint*, so its natural target is the half-shifted Hurwitz zeta. Before the figures, we state the algebra that makes this target legitimate, which is the single most likely reader sticking point.

5.1 Why the half-integers reproduce ζ's own zeros

A fair worry: the literal Dirichlet series is summed over the counting numbers $1, 2, 3, \ldots$ (that is the $\alpha = 1$ warp of Section 6), so why should a *half-integer* object reproduce ζ's zeros and their migration at all? Because the half-integer object is not a different beast that merely mimics ζ; it *contains* ζ *as a factor*, by an exact identity. The half-integers are the odd integers rescaled by $\frac{1}{2}$:

$$\zeta(s, \tfrac{1}{2}) = \sum_{n \geq 0} (n + \tfrac{1}{2})^{-s} = 2^s \sum_{n \geq 0} (2n+1)^{-s} = 2^s \sum_{m \text{ odd}} m^{-s}, \tag{12}$$

and "sum over the odds" is "all integers minus the evens":

$$\sum_{m \text{ odd}} m^{-s} = \zeta(s) - 2^{-s}\zeta(s) = (1 - 2^{-s})\zeta(s), \tag{13}$$

so that

$$\boxed{\zeta(s, \tfrac{1}{2}) = 2^s(1 - 2^{-s})\zeta(s) = (2^s - 1)\,\zeta(s).} \tag{14}$$

This is *algebra, not a limit* (it is the Hurwitz half-shift; see 11, 16 and 9, §25.11). Summing over the half-integers equals summing over the counting numbers times the explicit entire factor $2^s - 1$. The zeros therefore simply split,

$$\{\text{zeros of } (2^s - 1)\zeta(s)\} \;=\; \underbrace{\{\zeta\text{'s zeros on } \sigma = \tfrac{1}{2}\}}_{\zeta\text{'s own, identically}} \cup \underbrace{\{\, s : 2^s = 1 \,\}}_{\sigma = 0,\ t = 2\pi k/\ln 2}, \tag{15}$$

since $2^s = e^{s \ln 2} = 1$ forces $\sigma = 0$ and $t = 2\pi k/\ln 2$. So the $\sigma = \frac{1}{2}$ family that the warp migrates onto *is* ζ's genuine non-trivial zero set, identically, and not a look-alike.

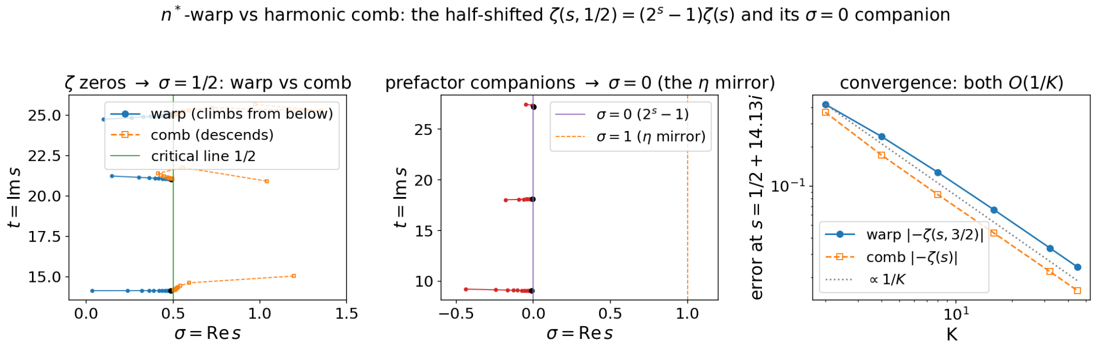

Figure 4: The nonlinear midpoint warp vs. the additive comb. The warp collapses each unit cell onto its midpoint, so it targets $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$ on the nose: ζ's zeros climb onto $\sigma = \frac{1}{2}$ *from below*, and the $2^s - 1$ companion migrates onto $\sigma = 0$ (the functional-equation mirror of η's $\sigma = 1$ comb). The two routes reach the critical line by *opposite* approaches, a contrast the rate law (Section 7) explains quantitatively.

Remark 1. The half-integer lattice carries the same prime content as ζ for every odd prime and isolates only $p = 2$ (the pulled-out Euler factor $1 - 2^{-s}$). That isolated factor reappears as the $2^s - 1$ companion on $\sigma = 0$, the functional-equation mirror of η's $\sigma = 1$ comb (Section 9) at the same heights. Using the counting numbers directly ($\alpha = 1$) would give plain $\zeta(s)$ and *only* the $\sigma = \frac{1}{2}$ family; the half-shift gives the *same* critical zeros plus that explicit, fully understood $\sigma = 0$ bonus, and does so from the pristine zeroless endpoint (1).

5.2 The warp and its zeros

With (14) in hand, the warp's behavior is clean (Fig. 4). The genuine ζ zeros climb onto $\sigma = \frac{1}{2}$ but, unlike the comb's zeros (born high, near $\sigma \approx 1.2$, and then descending), the warp's zeros are born *low* (near $\sigma \approx 0$) and climb *from below*. A companion family, the zeros of the $2^s - 1$ prefactor at $t = 2\pi k/\ln 2$, migrates onto $\sigma = 0$. Below $T = 40$ we measure a clean bijection: six zeros on $\sigma = \frac{1}{2}$ and four on $\sigma = 0$, matching the predicted counts from (15).

Observation 2 (Birth and migration are successive stages, not alternatives)**.** For the half-shift phase $\alpha = \frac{1}{2}$ the $K = 0$ warp object is the zeroless integral (1), $1/(s-1)$, with no non-trivial zeros at all, so at the start there is nothing to rearrange. The zeros are therefore *created* as harmonics are restored, and only afterwards do they move. This matches the integer phase $\alpha = 1$, which shares the same zeroless $K = 0$ endpoint; it is unlike the η side of Section 4, whose $K = 0$ ground string already carries zeros that then split. The lowest zeros appear already at the first harmonic, $K = 1$, and higher ones only as more harmonics are added (quantified by the birth-K law of Section 7). So for $\alpha = \frac{1}{2}$ both families are genuinely born out of the zeroless endpoint: the $\sigma = \frac{1}{2}$ family (the genuine ζ zeros of (15)) appears low, near $\sigma \approx 0$, and then migrates up onto $\sigma = \frac{1}{2}$, while the $\sigma = 0$ companions appear and settle onto $\sigma = 0$. The word "born" thus names the creation of a zero out of a zeroless endpoint, and "migrate" the subsequent motion onto a line; they describe two stages of one trajectory rather than a choice between them. That creation, not a shuffling of some fixed pre-existing set, is what sets the construction apart from the de Bruijn–Newman flow (Section 10).

5.3 The warp coordinate, visualized

Figure 5 pictures the warp itself rather than its zeros. The warp coordinate $n^*(x) = x + \varphi_K(x)$ starts as the linear x at $K = 0$ and, as harmonics are restored, bends into the midpoint staircase

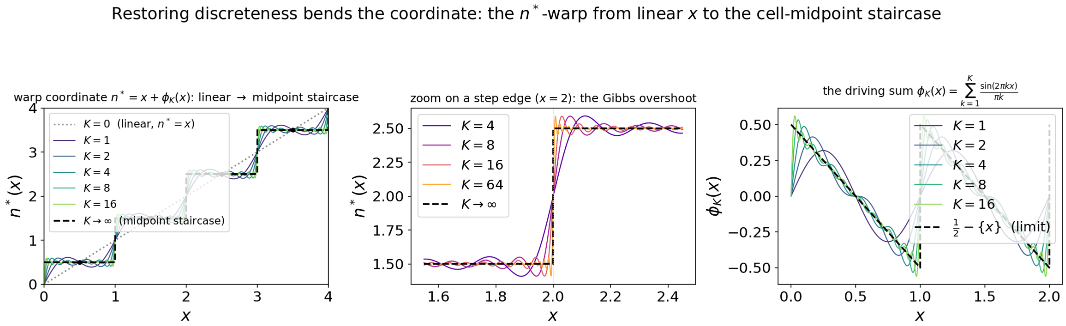

Figure 5: The warp coordinate $n^*(x) = x + \varphi_K(x)$ bending from the linear x ($K = 0$) into the midpoint staircase $\lfloor x\rfloor + \frac{1}{2}$ as Fourier harmonics are restored. Panel (b): the ∼9% Gibbs overshoot at the step edges, which narrows but never vanishes, the source of the warp's non-elementary rate constant.

	steps land on	$K = 0$ endpoint	$K \to \infty$ target
$\alpha = \frac{1}{2}$ (midpoint, headline)	half-integers $\frac{1}{2}, \frac{3}{2}, \ldots$	x (pristine $1/(s-1)$)	$(2^s - 1)\zeta(s) = \zeta(s, \frac{1}{2})$
$\alpha = 1$ (integer)	counting numbers $1, 2, 3, \ldots$	$x + \frac{1}{2}$ (DC offset on)	$\zeta(s, 1) = \zeta(s)$

Table 1: The trade-off: a zero-mean periodic correction added to x can only ever reach $\lfloor x\rfloor + \frac{1}{2}$. Landing on the integers *requires* the non-trigonometric constant $\frac{1}{2}$, so at $K = 0$ one is already at $x+\frac{1}{2}$, no longer the bland trig-free x. One can have a pure-x start *or* integer steps, not both.

$\lfloor x\rfloor + \frac{1}{2}$, collapsing each unit cell onto its midpoint. The cell midpoints $m + \frac{1}{2}$ are exact fixed points at every K. Because the underlying sawtooth has a unit jump at each integer, the partial sums carry the classic ∼9% Gibbs overshoot at the step edges, which narrows but never vanishes. That never-vanishing Gibbs boundary layer is precisely why the warp's rate constant is a non-elementary sine-integral rather than the comb's clean $s/2\pi^2$ (Section 7).

5.4 Half-integers vs. counting numbers: why $\alpha = \frac{1}{2}$ is the headline

Why does the staircase land on the half-integers rather than the counting numbers? Both are reachable: they are the $\alpha = \frac{1}{2}$ and $\alpha = 1$ cases of the general-phase warp (Section 6), whose cells collapse onto $\lfloor x\rfloor+\alpha$ via the DC-shifted displacement $\varphi_K^{(\alpha)}(x) = (\alpha-\frac{1}{2})+\varphi_K(x) \to \alpha-\{x\}$. Figure 6 puts them side by side. The obstruction to having both is the constant ($k = 0$, DC) Fourier coefficient: the $\alpha = \frac{1}{2}$ displacement is the *pure* sawtooth φ_K (zero mean), while $\alpha = 1$ carries a constant $+\frac{1}{2}$.

This is why the bridge's headline is $\alpha = \frac{1}{2}$: it is the one phase that keeps the pristine zeroless endpoint (1) *and* lands on a clean line, and that line is the richer half-shifted $(2^s-1)\zeta(s)$, whose $\sigma = \frac{1}{2}$ zeros are ζ's own (Section 5.1) and which also carries the $\sigma = 0$ companion. Choosing $\alpha = 1$ recovers plain $\zeta(s)$ and the literal counting numbers, but at the cost of the pristine endpoint *and* the companion line.

5.5 Warping the η integral: why $\alpha = \frac{1}{2}$ is a ζ-only luxury

The warp has so far acted only on the *bland* integrand x^{-s}. It is natural to ask what it does to the bridge's *structured* endpoint, the continuous-η integral (3): warp the whole integrand,

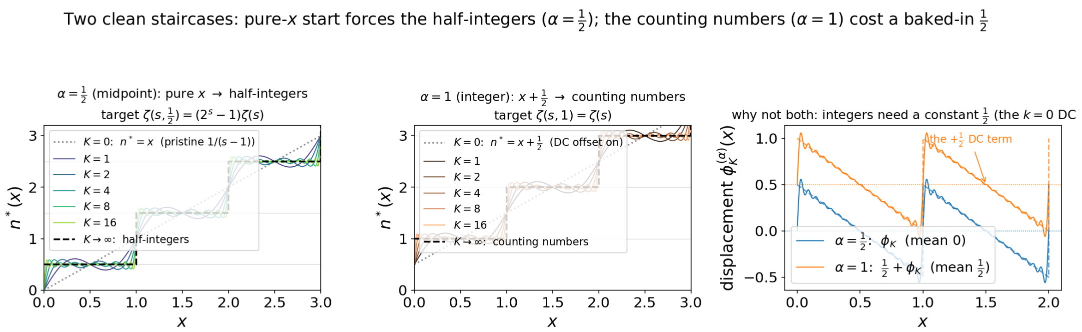

Figure 6: Two clean staircases: $\alpha = \frac{1}{2}$ (half-integers) and $\alpha = 1$ (counting numbers). Panel (c) shows the obstruction: the $\alpha = 1$ displacement carries a constant DC term while $\alpha = \frac{1}{2}$ does not. The half-integer midpoint is the *unique* clean staircase reachable from a pure-x start by pure trigonometric corrections, which is why it is the bridge's headline phase.

sign-carrier and all,

$$\int_1^\infty \cos(\pi n^*)\,(n^*)^{-s}\,\mathrm{d}x, \qquad n^* = x + (\alpha - \tfrac{1}{2}) + \varphi_K(x). \tag{16}$$

The period-by-period collapse is $\sum_{m\geq 1}(-1)^m \int_0^1 \cos(\pi g)(m+g)^{-s}\,\mathrm{d}u$ with $g(u) = u + (\alpha - \frac{1}{2}) + \varphi_K(u)$, and as $K \to \infty$ each cell lands on $\cos(\pi(m+\alpha))\,(m+\alpha)^{-s}$. The phase now matters in a way it did not for ζ:

- **The midpoint phase $\alpha = \frac{1}{2}$ is annihilated.** $\cos(\pi(m+\frac{1}{2})) = 0$ identically: the midpoint lattice $\{m+\frac{1}{2}\}$ *is the zero set of the η sign-carrier* $\cos(\pi x)$, so (16) collapses to 0 (numerically as $O(1/K)$, the finite-K residual being the Gibbs smoothing of those zeros). The elegant half-shift of Section 5 has *nothing to land on.*
- **Only the integer phase $\alpha = 1$ survives.** There $\cos(\pi(m+1)) = (-1)^{m+1}$, so (16) $\to -\eta(s) + 1$; restoring the omitted $n = 1$ cell $\cos\pi \cdot 1^{-s} = -1$ (the load-bearing lower-endpoint term, the η-form of the warp's $+2^s$) completes it to $\eta(s) = (1-2^{1-s})\zeta(s)$. Its zeros are the genuine ζ zeros on $\sigma = \frac{1}{2}$ together with the $1 - 2^{1-s}$ comb on $\sigma = 1$: the *same* split the additive comb produced in Section 4, now reached by the independent warp mechanism (Fig. 7), at the same $O(1/K)$ rate.

This sharpens Section 5.4: ζ could afford $\alpha = \frac{1}{2}$ (keeping the pristine zeroless endpoint *and* a clean line) because its integrand x^{-s} has no zeros on the half-integer lattice. The η integrand does, so η is forced onto $\alpha = 1$ and pays the DC-offset endpoint that ζ avoided. The half-shift is special for ζ because the discreteness it samples carries no alternating sign.

6 The general phase and the $P \cdot L$ dichotomy

Generalizing the warp to pin cells to an arbitrary phase $n + \alpha$, $\alpha \in (0, 1)$, the completed target is the Hurwitz zeta $\zeta(s, \alpha)$. Sweeping α and watching where the zeros go exhibits the Saias–Weingartner $P \cdot L$ dichotomy [30]: clean vertical zero strings appear *only* at $\alpha \in \{0, \frac{1}{2}, 1\}$, the phases at which $\zeta(s, \alpha)$ factors as a polynomial in prime powers times a single Dirichlet L-function. (For $\alpha = \frac{1}{2}$ this is (14); in the language of 30 the relevant identity is $(1-2^{-s})\zeta(s) = P(s)\,L(s, \chi_0)$, with 2^s entire and nonvanishing.) Every *generic* α gives a Davenport–Heilbronn scattered set, with zeros off any vertical line and even into $\sigma > 1$ [6, 8, 30]; see also the value-distribution results for algebraic-irrational α in Sourmelidis and Steuding [33].

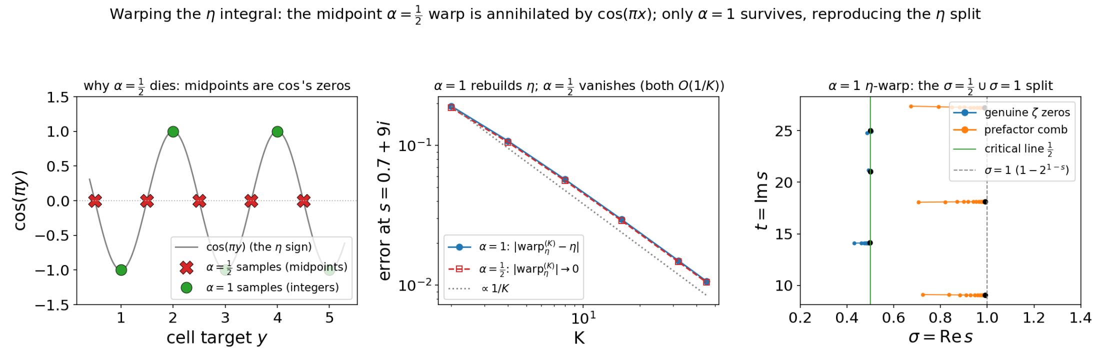

Figure 7: Warping the η integral (16). *Left:* the obstruction. The midpoints $m+\frac{1}{2}$ that the $\alpha=\frac{1}{2}$ warp samples are the zeros of the η sign-carrier $\cos(\pi x)$, while the integers m that $\alpha=1$ samples sit on its extrema ± 1. *Middle:* consequently $\alpha=1$ rebuilds η at $O(1/K)$ while the $\alpha=\frac{1}{2}$ warp vanishes. *Right:* the $\alpha=1$ η-warp's zeros migrate onto $\sigma=\frac{1}{2}$ (genuine ζ zeros) and $\sigma=1$ (the prefactor comb): the same split as the additive comb (Fig. 3), by a second, independent route.

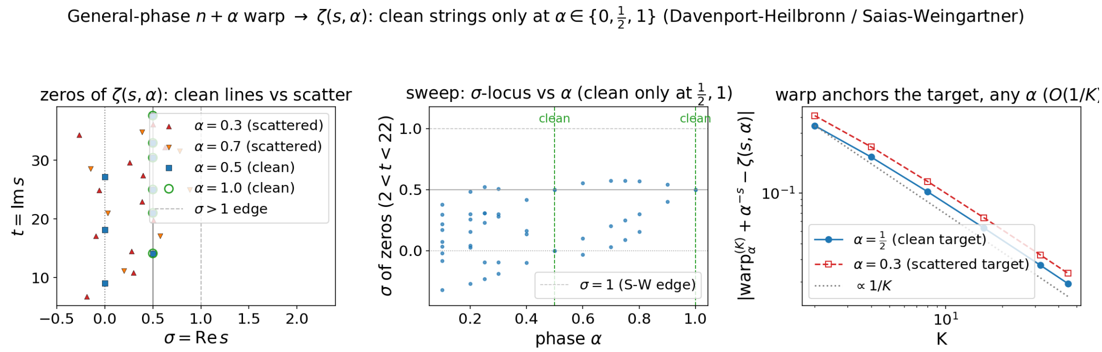

Figure 8: The general-phase $n+\alpha$ warp onto $\zeta(s,\alpha)$, exhibiting the Saias–Weingartner dichotomy: clean vertical strings only at $\alpha\in\{0,\frac{1}{2},1\}$; every generic α is Davenport–Heilbronn scattered off-line. This locates the half-shift of Section 5 inside a one-parameter family and explains why $\frac{1}{2}$ is *special* rather than merely chosen.

This locates the half-shift of Section 5 inside a one-parameter family and explains why $\frac{1}{2}$ is special rather than merely chosen: it is one of the three privileged phases. The "clean lines only at $\{0,\frac{1}{2},1\}$" picture is the load-bearing control that keeps the half-shift route honest.

7 The unified $O(1/K)$ rate law

The synthesis is that one mechanism sits behind every route. All routes truncate the *same* sawtooth (2), whose retained tail has power

$$\sum_{k>K}\frac{1}{(\pi k)^2}\sim\frac{1}{\pi^2 K},\tag{17}$$

so every route converges as $O(1/K)$. A single zero-displacement law,

$$s_K-\rho\;\sim\;\frac{a_1(\rho)}{K},\tag{18}$$

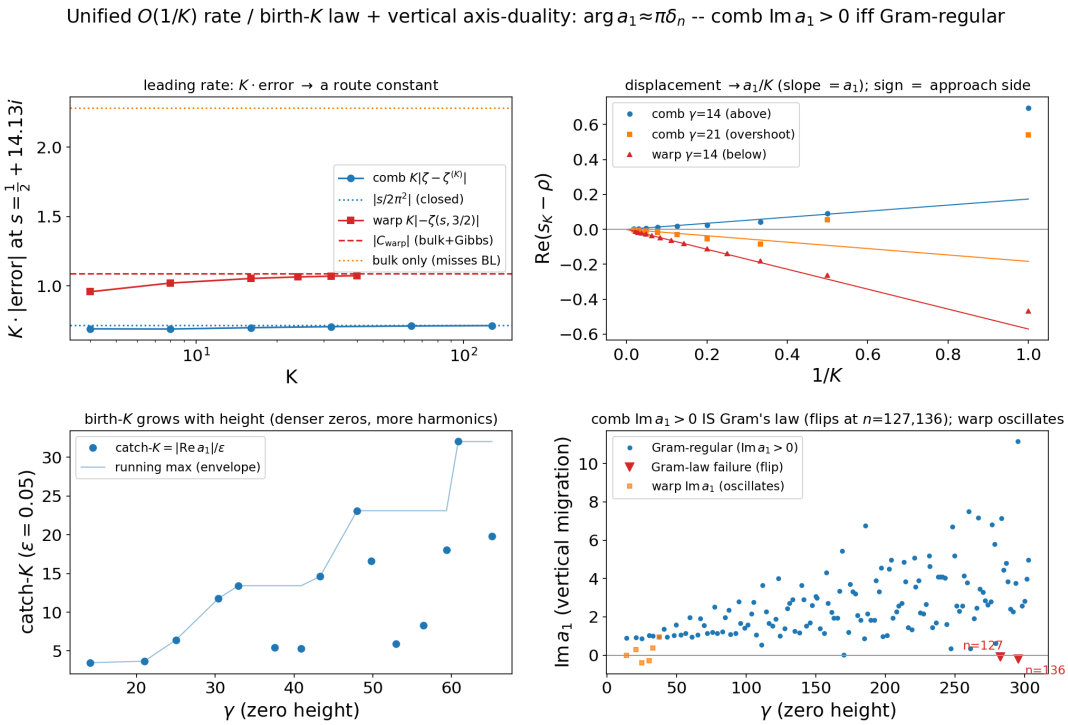

Figure 9: The unified $O(1/K)$ rate / birth-K law and the vertical axis-duality. All routes truncate the same sawtooth, so all converge as $O(1/K)$ (17); the displacement constant $a_1(\rho)$ of (18) controls approach side ($\mathrm{sign}\,\mathrm{Re}\, a_1$), birth order ($K_{\mathrm{catch}} = |\mathrm{Re}\, a_1|/\varepsilon$), and the vertical migration ($\mathrm{Im}\, a_1$), whose sign tracks Gram's law.

covers all five zero families (the ζ comb, the η split onto $\sigma = \frac{1}{2}$ and $\sigma = 1$, and the warp's $\sigma = \frac{1}{2}$ and $\sigma = 0$ families), with a route- and zero-dependent complex displacement constant $a_1(\rho)$. The structure of a_1 carries the qualitative behavior:

- **Approach side** $= \mathrm{sign}(\mathrm{Re}\, a_1)$**.** The real part of a_1 fixes which side the zero approaches its line from, which is why the comb descends (and can overshoot) while the warp climbs from below and never overshoots; overshoot occurs precisely when $\mathrm{Re}\, a_1 < 0$.
- **Birth-K law.** The first K that resolves a given zero to tolerance ε is $K_{\mathrm{catch}} = |\mathrm{Re}\, a_1|/\varepsilon$. This ties the birth order of zeros to the Montgomery a-values $1/|\zeta'(\rho)|$.
- **Vertical migration = Gram's law.** The imaginary part $\mathrm{Im}\, a_1$ gives the vertical (along-t) migration. Its sign tracks Gram's law numerically: it first flips at the classical Gram-law failures (zeros $n = 127, 136, 196$).

This is where the package stops being several separate demonstrations and becomes one law: a single rate constant and displacement coefficient, route by route, plus a Gram's-law coupling on the vertical axis. It is the synthesis around which the whole study is organized.

7.1 Stability of the interpolants: no abscissa, height is the cost

The rate law also settles a question the Dirichlet series provokes: how far into the plane do the truncated objects remain valid? The series $\sum n^{-s}$ converges only for $\mathrm{Re}\, s > 1$ and its alternating cousin only for $\mathrm{Re}\, s > 0$, so one might expect $\zeta^{(K)}$ to inherit a barrier. It does not. Each $\zeta^{(K)}$ is *not* a partial sum of a Dirichlet series but $1/(s-1) + \frac{1}{2}$ plus finitely many harmonics, every one of which is an incomplete-gamma closed form, entire in s for its fixed nonzero argument.

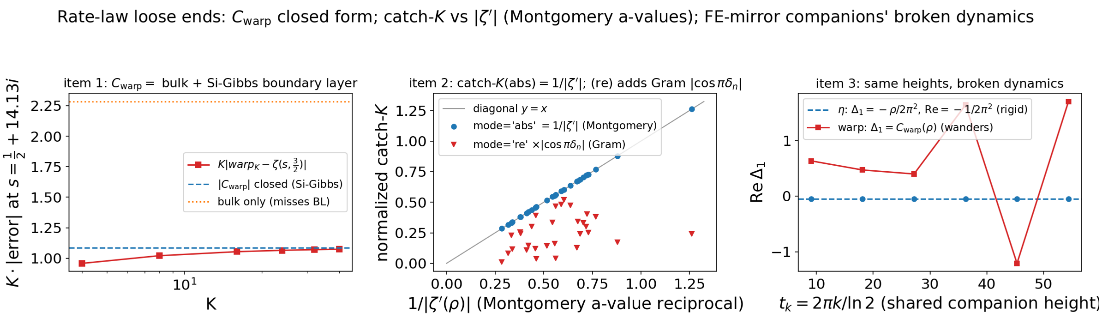

Figure 10: Three loose ends of the rate law: (i) a closed form for the warp constant C_{warp} as a Gibbs boundary-layer (sine-integral) integral; (ii) the tie of the birth threshold K_{catch} to the Montgomery a-values $1/|\zeta'(\rho)|$; and (iii) the fact that the $\sigma = 1/\sigma = 0$ companions share heights but *not* migration dynamics.

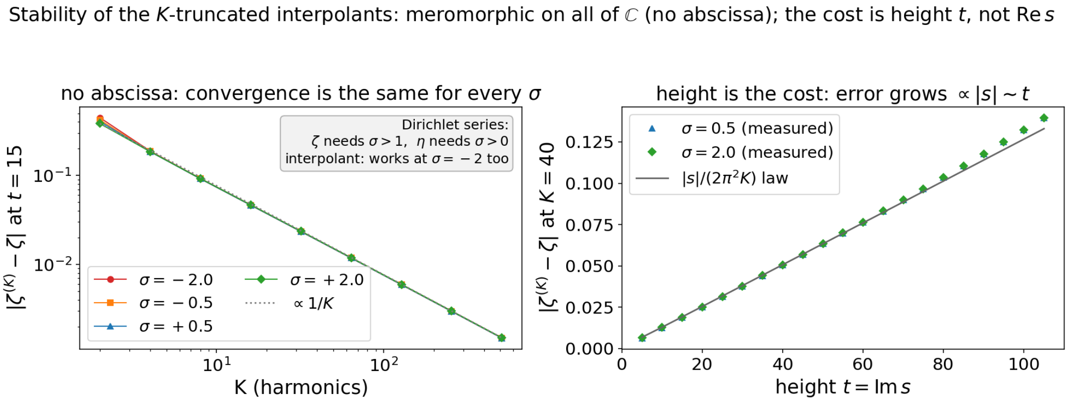

Figure 11: Where the K-truncated interpolants are stable. *Left:* the convergence error $|\zeta^{(K)} - \zeta|$ vs. K collapses onto one $O(1/K)$ line for every σ, including $\sigma < 0$ where both Dirichlet series diverge, so there is no abscissa. *Right:* at fixed K the error grows linearly with height t, tracking the analytic rate constant $|s|/(2\pi^2 K)$: the cost is height, not Re s.

So every finite-K interpolant is meromorphic on all of $\mathbb{C}$, with the single pole at $s = 1$ and *no abscissa of convergence at all*: the zeroless endpoint $1/(s-1)$ already shows this, and each harmonic preserves it. The interpolant is its own analytic continuation, strictly better behaved than either series.

The cost of reaching deep into the plane is therefore not a half-plane boundary but *height*, and two facts make this precise (Fig. 11). First, Re s is essentially free: $\zeta^{(K)} \to \zeta$ at the *same* $O(1/K)$ rate at $\sigma = -2$ as at $\sigma = 2$, the convergence curves for $\sigma \in \{-2, -\frac{1}{2}, \frac{1}{2}, 2\}$ lying on top of one another. Second, the cost is set by $t = \mathrm{Im}\, s$ through the comb rate constant $\Delta_1 = s/(2\pi^2)$ of Section 7 (the sawtooth tail (17)): the fixed-K error tracks $|s|/(2\pi^2 K)$, growing linearly with height. This is the convergence-side face of the birth-K law (higher, denser zeros need proportionally more harmonics), and it depends on t, not on any vertical line in σ.

The only genuine limit is finite precision, and it too is governed by height: at large t the moment evaluators carry $e^{\pi t/2}$-scale cancellations, so a fixed working precision eventually floors (we observe this near $|\mathrm{Im}\, s| \approx 110$ and remove it by scaling precision, term count and quadrature degree with $|\mathrm{Im}\, s|$). But the $O(1/K)$ convergence is slow enough that at moderate height this floor sits far below the achievable error and the descent stays clean; in exact arithmetic there is no divergence at any K, and more harmonics always help.

8 The trivial zeros: a rising tide on the negative axis

Everything so far has tracked the *non-trivial* zeros. But ζ also carries the *trivial* zeros at $s = -2, -4, -6, \ldots$, and the bland endpoint $1/(s-1) + \frac{1}{2}$ has none of them (its only real zero is at $s = -1$). They must therefore be born somewhere along the interpolation as well, and the stability result of Section 7.1 says exactly where we are allowed to look: because $\zeta^{(K)} \to \zeta$ with no abscissa, converging at $\sigma = -2$ as readily as at $\sigma = +2$, the negative real axis is fair game. Following it gives a clean account of the trivial string. On that axis every harmonic in (7) is real, so $\zeta^{(K)}$ is a real function whose trivial zeros are honest sign changes, and the leading $O(1/K)$ tail (17) acts as a *rising tide*: a positive, uniform lift

$$\zeta^{(K)}(s) \;\approx\; \zeta(s) + \frac{|s|}{2\pi^2 K}, \qquad s \text{ real},\; s < 0, \tag{19}$$

governed by the *same* comb rate constant $|s|/(2\pi^2 K)$ that set the height cost in Section 7.1. The genuine ζ oscillates about zero on the negative axis, with the trivial zeros as its crossings; the tide fills in the shallow dips where $\zeta < 0$ and erases the crossing pairs that bound them, and as K grows the tide recedes and uncovers the zeros (Fig. 12a). Three things follow.

The first trivial zero -2 is inherited, not born. The dip on $(-2, 0)$ is bounded on the right by the pole side, not by a zero, so the endpoint's lone real zero (the $s = -1$ of $1/(s-1) + \frac{1}{2}$) is never destroyed: it simply slides leftward and limits onto -2, as $-2 + a_1/K$ with $a_1 = -(-2)/(2\pi^2\zeta'(-2)) \approx 3.327$. The bland integral's single zero is the seed of the entire trivial string.

The rest are born as conjugate pairs that pinch onto the axis. Below its birth-K a pair is not absent from the plane: it is a complex-conjugate pair sitting *off* the real axis, near an odd-integer midpoint $-(4j+1)$. As K grows the pair descends, *pinches* onto $\operatorname{Im} s = 0$, and splits, one zero sliding right toward $-4j$ and one left toward $-(4j+2)$ (Fig. 12b). For example the $\{-4, -6\}$ pair runs $-4.75 + 0.88\mathrm{i}$ at $K = 30$, pinches near -4.78 at $K \approx 62$, then splits to -4 and -6.

Observation 3 (The trivial zeros mirror the non-trivial story)**.** This is the non-trivial picture rotated onto the real axis and run in pairs. The non-trivial zeros are *born off their locus and migrate onto it*; so are these, except the locus is the negative real axis and the approach is a conjugate pair pinching onto it from above and below. The one feature with no non-trivial analogue is the *birth event* itself: the non-trivial families are present already at $K = 1$ and only migrate thereafter, whereas a trivial pair must first pinch onto the axis before it exists as two real zeros.

Deep valleys are born first. A pair clears the tide once $|s_{\mathrm{mid}}|/(2\pi^2 K)$ drops below the valley depth $|\zeta(s_{\mathrm{mid}})|$, i.e. at

$$K_{\mathrm{pinch}} \;\approx\; \frac{|s_{\mathrm{mid}}|}{2\pi^2\,|\zeta(s_{\mathrm{mid}})|}, \qquad s_{\mathrm{mid}} = -(4j+1). \tag{20}$$

Since $|\zeta(-(2m+1))| = |B_{2m+2}|/(2m+2)$ grows *factorially*, the deep valleys are bottomless and their pairs are present already at $K \sim 1$, while the *shallowest* valley is the first one (depth $|\zeta(-5)| \approx 0.004$), so the $\{-4, -6\}$ pair is the *last* to clear: not born until $K \approx 62$ (measured), against $K \approx 64$ from (20), whereas the deeper $\{-12, -14\}$ pair is already born by $K \approx 8$ (Fig. 12c). The negative axis thus fills in from both ends, and the hardest trivial zeros to resolve are -4 and -6.

This account needs no new machinery. The displacement law (18) applies verbatim with target ζ and $\rho = -2n$: the trivial displacement coefficient is identically the comb's $a_1(-2n)$ to

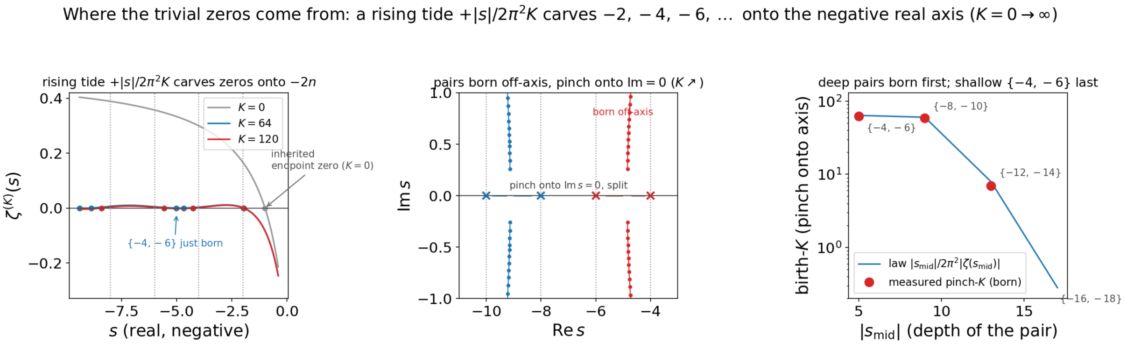

Figure 12: Where the trivial zeros come from. *(a)* On the negative real axis the $O(1/K)$ tail is a rising tide $+|s|/(2\pi^2 K)$ (19) that lifts $\zeta^{(K)}$ above zero and recedes as K grows, uncovering the trivial zeros at $-2, -4, -6, \ldots$; the inherited endpoint zero ($K = 0$) is marked. *(b)* Each born pair starts as a complex-conjugate pair off the axis, descends as K grows, pinches onto $\operatorname{Im} s = 0$, and splits toward adjacent even integers. *(c)* The birth-K law (20): valley depth grows factorially, so deep pairs are born first and the shallow $\{-4, -6\}$ pair is born last (measured points against the law).

full precision. What is genuinely new on this axis is only the birth event of Observation 3—the conjugate-pair pinch and its valley-depth birth-K law (20)—which the non-trivial families never undergo. With it the bridge accounts for *all* of ζ's zeros: the condensation of the non-trivial ones onto their critical lines, and the birth of the trivial ones out of a zeroless endpoint.

9 The two-component η spectrum

Projected onto the t-axis, the η zero set is a *rigid crystalline comb* ($\sigma = 1$ teeth at $t = 2\pi k/\ln 2$, perfectly periodic) *superposed* with a *GUE-random sequence* (the $\sigma = \frac{1}{2}$ ζ zeros) [22, 24]. Figure 13 applies three lenses:

1. **One-point packing.** There are $\log_2(t/2\pi)$ zeros per comb gap, equidistributed.

2. **Two-point number variance.** In the smooth bulk both lenses say the components behave as an *independent superposition.*

3. **Resonance at the comb frequency.** The third lens shows they are *not* independent. The comb teeth sit at the $p = 2$ prime-power frequencies $m \ln 2$, and by the explicit formula the ζ-zero density carries an oscillation at every prime-power frequency, so probing the zeros at the comb frequency *resonates*: the measured $\langle \cos(m\gamma \ln 2)\rangle$ matches the $p = 2^m$ explicit-formula coefficient to $\sim$4 digits, with a negative sign (the comb teeth sit at density *minima*).

   The $\sigma = 1$ comb is, in this sense, the "$p = 2$ ghost" of the zeros' own prime structure.

Remark 2 (Honest caveat)**.** The appearance of $\ln 2$ in the zeros' number variance is *not* new. Berry [3], made rigorous by Lugar, Milinovich and Quesada-Herrera [20], already places prime-power log-frequencies in the zeta-zero number variance in a regime where the GUE prediction fails, and $\ln 2$ is the *single lowest* such frequency (note $\ln 3 < \ln 4 = 2\ln 2$, so $\ln 2$ is lowest but the band is not simply the multiples of $\ln 2$). What is ours here is only the comb$\uplus$GUE *superposition* model, the *single-prime isolation* ($p = 2$), and the *eta tie-in* (relating the $\sigma = 1$ eta comb to that lattice), not the $\ln 2$ frequency itself. See Section 10.

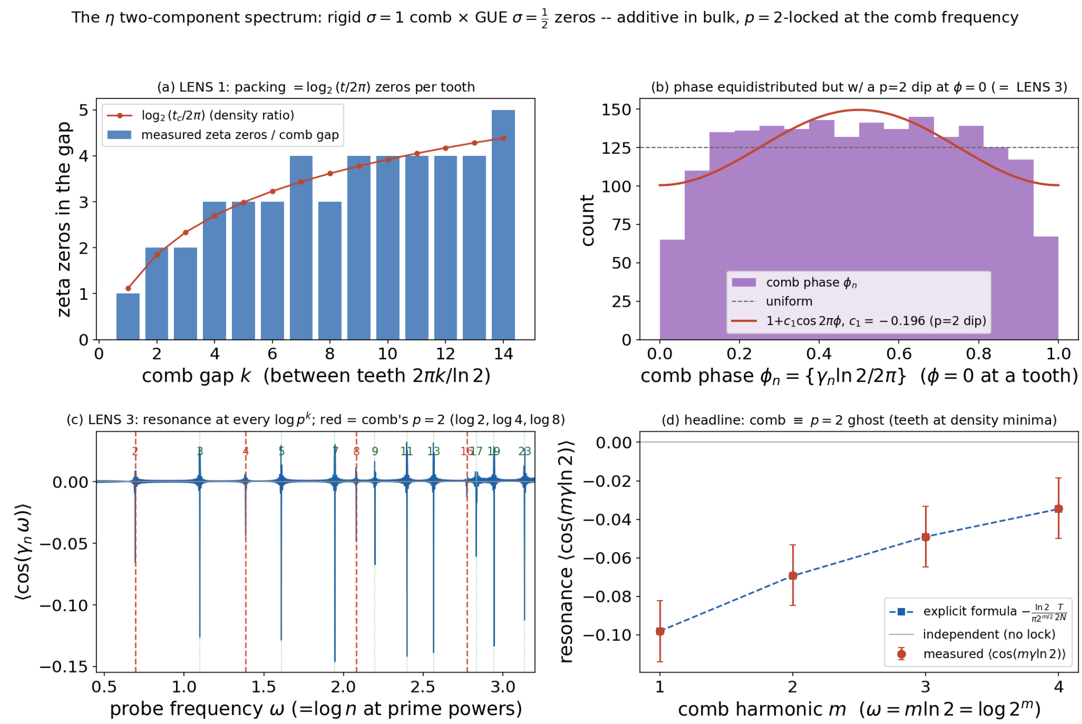

Figure 13: The η zeros as a two-component "crystal $\times$ GUE" spectrum: a rigid $\sigma = 1$ comb at $t = 2\pi k/\ln 2$ superposed on the $\sigma = \frac{1}{2}$ GUE ζ-zeros. The comb frequencies $m \ln 2$ are the $p = 2$ prime-power frequencies; probing the zeros there resonates with the explicit-formula coefficient (the "$p = 2$ ghost").

10 Relation to prior work

The endpoints and most ingredients of this study are classical; we cite them rather than claim them. We have made two adversarial passes through the literature, and we foreground the closest prior-art results explicitly so a reader can locate our residual contribution precisely.

10.1 The deform-and-track lineage and its direct ancestor

Deforming a Dirichlet series by a parameter and following its zeros is a program roughly half a century old, and its direct ancestor is Fornberg and Kölbig [10]: taking the polylogarithm $F(x, s) = \sum_{k \geq 1} x^k k^{-s}$ (with $F(1, s) = \zeta(s)$ and $F(-1, s) = -\eta(s)$), they sweep the argument x across $(-1, 1)$ and numerically track the complex zeros in the s-plane. The distinction from the present construction is the one we lead with: on their argument knob the moving zeros *graze* the critical line and are then absorbed into the pole emerging at $x = 1$, so the approach is transient; the harmonic-count knob here, off a genuinely zeroless endpoint, instead makes the zeros *converge* onto $\sigma = \frac{1}{2}$ at the $O(1/K)$ rate of Section 7 and stay. This is a sharper statement than "we use a different knob": it is a knob on which the convergence the classical deformation set out toward actually occurs. The closest *published* analogue in the same spirit is the Hurwitz $\zeta(s, \alpha)$ zero-trajectory study of Garunkštis and Steuding [11], which deforms the shift α and calls a zero "stable" when its trajectory begins and ends on $\sigma = \frac{1}{2}$; the reverse-direction mirror is the Davenport–Heilbronn homotopy of Balanzario and Sánchez-Ortiz [1], in which zeros start *on* the line and are driven *off* it as the periodic structure is switched on. Appendix A reproduces the polylogarithm ancestor and the Hurwitz/Lerch analogues, sets against them the reverse-direction Epstein foil of Travěnec and Šamaj [36], and runs the bridge's own knob forward on a primitive Dirichlet L-function—four families beyond ζ/η, organized by a directional taxonomy.

10.2 Cite, do not claim

- The half-shift $(2^s-1)\zeta(s)=\zeta(s,\frac{1}{2})$, its $\sigma=0$ companion at $t=2\pi k/\ln 2$, and the migration of Hurwitz-type zeros onto $\sigma=\frac{1}{2}$ as a parameter varies: Garunkštis and Steuding [11], Gonek [16]; see also Garunkštis and Tamošiūnas [13] for the periodic-Hurwitz Riemann–von Mangoldt counting law.
- The $\alpha\in\{0,\frac{1}{2},1\}$ "clean lines only" dichotomy: Saias and Weingartner [30]; generic off-line $\zeta(s,\alpha)$ zeros: Cassels [6], Davenport and Heilbronn [8].
- The $\sigma=1$ eta comb at $t=2\pi k/\ln 2$: Sondow [32]; tabulated by Beliakov and Matiyasevich [2]; 9, §25.2.3. These points appear earlier still, in a deform-and-track setting, as the $x\to-1$ endpoint zeros of Fornberg and Kölbig [10, Eq. 23]; since the points are elementary (the zeros of $1-2^{1-s}$), this predates but does not displace Sondow's η-specific result.
- The closest "knob" cousins use partial-sum *length* N rather than the harmonic count K, and none start from a zeroless endpoint: Gonek and Ledoan [14], Gonek and Montgomery [15]; see also Platt and Trudgian [25]. Zeros of combinations of Euler products for $\sigma>1$: Booker and Thorne [5], Righetti [28].

10.3 The deflator to foreground (the $p=2$ ghost)

As stated in Remark 2, the strongest prior art for the two-component reading is Berry [3], proved by Lugar et al. [20]: the number variance $\Sigma^2(L)$ of the Riemann zeros has a prime-controlled regime in which the GUE prediction explicitly fails, governed by a von Mangoldt cosine sum at frequencies $\ln n$ whose single smallest is $\ln 2$. That work treats the *single* zeta-zero spectrum; it does not treat the Dirichlet-η / partial-sum zeros, does not build the comb $\uplus$ GUE two-spectrum superposition statistics, does not isolate a single prime, and does not tie the $\sigma=1$ eta comb at $t=2\pi k/\ln 2$ to the $p=2$ lattice via the explicit formula. Those four are the surviving contribution of Section 9. Related "quasicrystal" readings of the zeros (the zero-spacing distributions of differenced L-functions [19], and prime-power Bragg peaks in the zeros' structure factor [21, 31]) supply the "crystal" half as a single spectrum but not the superposition model; Morán Ledezma [23] treats the explicit formula as Bohr-space spectral measures (a value-distribution statement, not a zero-statistics one).

10.4 Contrast with the de Bruijn–Newman heat flow

Our deformation should not be confused with the de Bruijn–Newman heat flow, though the two sides of the bridge stand in different relations to it. In the heat-flow picture the family $H_t=\mathrm{e}^{t\Delta}H_0$ is, for *every* value of the flow parameter t, an entire function of order one carrying the full infinite complement of ξ-type zeros at Weyl density; the backward heat flow merely transports these already-existing zeros (by a Calogero–Moser-type ODE) onto or off the real axis, and the de Bruijn–Newman constant Λ (now known to satisfy $0\le\Lambda\le 0.22$, with RH $\iff \Lambda=0$ after 26, 29) marks the threshold at which an existing configuration has become entirely real. No zeros are created or destroyed; the question is purely *when* a fixed-cardinality configuration becomes all-real.

The ζ side: creation, not real-ification. On the ζ side the contrast is sharp. The harmonic-count knob K interpolates from a genuinely *zeroless* endpoint (the bare continuous integral (1), equivalently the $K=0$ warp object, which has no non-trivial zeros at all) to the full Dirichlet series, and the non-trivial zeros are *born* as K leaves 0 and only then migrate to the critical line (Observation 2). There is no fixed pre-existing configuration being rearranged, and no Λ-type threshold: this is creation-and-migration from a zeroless endpoint, not the real-ification of a pre-existing infinite zero set.

The η side: the nearer cousin, but still distinct. On the η side the analogy is closer, and we flag it rather than paper over it. The $K = 0$ ground string there is *not* zeroless: it is $-F(s) + \frac{1}{2}$, which already carries an infinite zero string at the full Riemann–von Mangoldt density (Observation 1); as $K \to \infty$ those zeros are *transported*, not created, in the same fixed-cardinality, density-preserving motion of a pre-existing infinite zero set that the heat-flow picture has. Three features nonetheless keep it distinct. First, the destination is a pair of *vertical* critical lines ($\sigma = \frac{1}{2}$ and $\sigma = 1$), not the real axis of a symmetrised ξ. Second, the motion is a density-conserving *split*: the single ground string is *sorted* onto two lines with the density bookkeeping of (11), a bifurcation driven by the arithmetic $1 - 2^{1-s}$ comb for which the heat flow (which keeps one configuration on one axis) has no analogue. Third, the deformation parameter is a harmonic count, the dynamics are not a Calogero–Moser gradient flow, and there is no value of K at which an all-line configuration is first achieved: the lines are reached only in the limit. So the bridge separates a genuine "born from nothing" mechanism (ζ) from a "transported and sorted" one (η); only the latter resembles the heat flow, and even then only in the weak sense that a pre-existing infinite zero set is moved by a parameter.

10.5 Contrast with Taylor-section and resummation routes

Our rate is algebraic, $O(1/K)$ (17). This distinguishes the construction from the Taylor sections of ξ, whose zeros converge *super-exponentially* [18], and from the Borel/resurgent resummation of the *entire* Euler–Maclaurin remainder [7], which retains no truncation parameter and births no zeros. The shared sawtooth tail (17) is what makes *our* family uniformly $O(1/K)$ across all routes.

11 Reproducibility

Every figure in this paper is produced by a driver of the same name in the accompanying repository, and every driver self-validates its identities to full precision before it plots, so each figure doubles as a numerical certificate. All seventeen figures—the thirteen in the body and the four of Appendix A—are regenerated together by a single command:

`python repro.py` (regenerates all figures); `pytest` (fast self-checks; `pytest -m slow` for the high-precision suite).

The drivers are pure `mpmath` except `eta_two_component.py`, which additionally reads a cached table of Riemann zeros. The code targets Python 3.9 and is released under the MIT license. Full setup instructions, a figure-by-figure walkthrough (`RESULTS.md`), and the annotated bibliography accompany the source at `https://github.com/bradleypmartin/dirichlet-bridge`.

Acknowledgments

This study was extracted as a self-contained arc from a larger research project. The author thanks Bengt Fornberg for flagging the program's direct ancestor—his own study of the complex zeros of the polylogarithm [10], whose zero-trajectory computation was run on the CDC 7600 at CERN in the mid-1970s, then among the fastest machines in the world, and is reproduced here (Appendix A) in seconds on a laptop half a century later. The novelty framing above rests on two adversarial literature passes; the author thanks the reviewers and tools that contributed to that vetting. Any errors of attribution are the author's own.

A Beyond ζ and η: four rhyming cases

The body used ζ and η as the two showcase test cases. This appendix records four *rhyming* cases beyond them: the program's direct ancestor (Appendix A.1), a genuine primitive Dirichlet L-function to which the bridge's own knob is applied (Appendix A.2), the closest published parameter-sweep analogues (Appendix A.3), and a reverse-direction foil (Appendix A.4). Each is a self-validating driver in the repository (Section 11), and each section is a faithful *reproduction* of prior art together with a *compare-and-contrast* against the bridge; none is a new theorem. The main technical arc stays ζ/η-only, and this material is deliberately supplementary.

A directional taxonomy. The organizing idea is a taxonomy of *where the zeros go* under a one-parameter deformation (Table 2). The half-century deform-and-track lineage (Section 10.1) is large, but every ancestor we found either deforms between two endpoints that *both* already carry infinitely many zeros—so no zero is *born*, and the trajectories merely wander on and off $\sigma = \frac{1}{2}$—or *births* zeros *off* the critical line and pushes them *away*. The bridge's motion is the only one whose arrow points *onto* the line from a zeroless start (or, for a mean-zero periodic coefficient, from a structured off-critical string).

Direction of the zeros	Family / knob	Closest sources
zeroless $\to$ **born** $\to$ **onto** $\frac{1}{2}$ *(bridge)*	$\zeta, \eta, L(s, \chi_4)$; harmonic count K	*this work* (Appendix A.2)
zero-rich $\to$ zero-rich; **wander on/off** $\frac{1}{2}$	polylog (arg x), Hurwitz (α), Lerch (λ, α)	[10–12]
born off $\frac{1}{2}$ $\to$ **migrate away**	Davenport–Heilbronn (homotopy), Epstein (dim. d)	[1, 36]

Table 2: The directional taxonomy of the deform-and-track lineage. The bridge's arrow (top row) is the only one that points *onto* $\sigma = \frac{1}{2}$ from a start that carries none of the destination's zeros.

A.1 The direct ancestor: the polylogarithm deformation

The program's direct ancestor is Fornberg and Kölbig [10], who study the complex zeros, in the s-plane, of the Jonquière (polylogarithm) function

$$F(x, s) = \sum_{k \geq 1} \frac{x^k}{k^s} \qquad (|x| < 1), \qquad F(1, s) = \zeta(s), \quad F(-1, s) = -\eta(s), \tag{21}$$

using the argument x as a deformation knob and tracking the zero trajectories as x sweeps $(-1, 1)$. The endpoints are exactly the bridge's two showcase objects. The driver `jonquiere_zeros.py` reproduces their half-century-old CDC 7600 computation on a laptop in seconds, recovering (Fig. 14) the two trajectory classes; the transient $\sigma = \frac{1}{2}$ *brush* of the first ζ zero $s_1 \approx \frac{1}{2} + 14.13\,\mathrm{i}$ and the *spiral* about it (FK Figs. 6–7); the $\sigma = 1$ log-2 comb at $1 + 2\pi \mathrm{i} m / \ln 2$ (FK Eq. 23); the trivial-zero trajectories running onto $-2m$ (FK §6); the argument-principle counts $Z(0.1) = 27$, $Z(-0.1) = 26$ (FK Eq. 25); and the Riemann–von Mangoldt counting constants $\alpha^+ = 0.00580756$, $\alpha^- = 0.01845107$ (FK Eqs. 39–40), all to the printed digits.

The contrast: a transient brush, not convergence. The distinction that motivates the whole bridge is visible in this reproduction. On FK's argument knob the moving zeros *do*

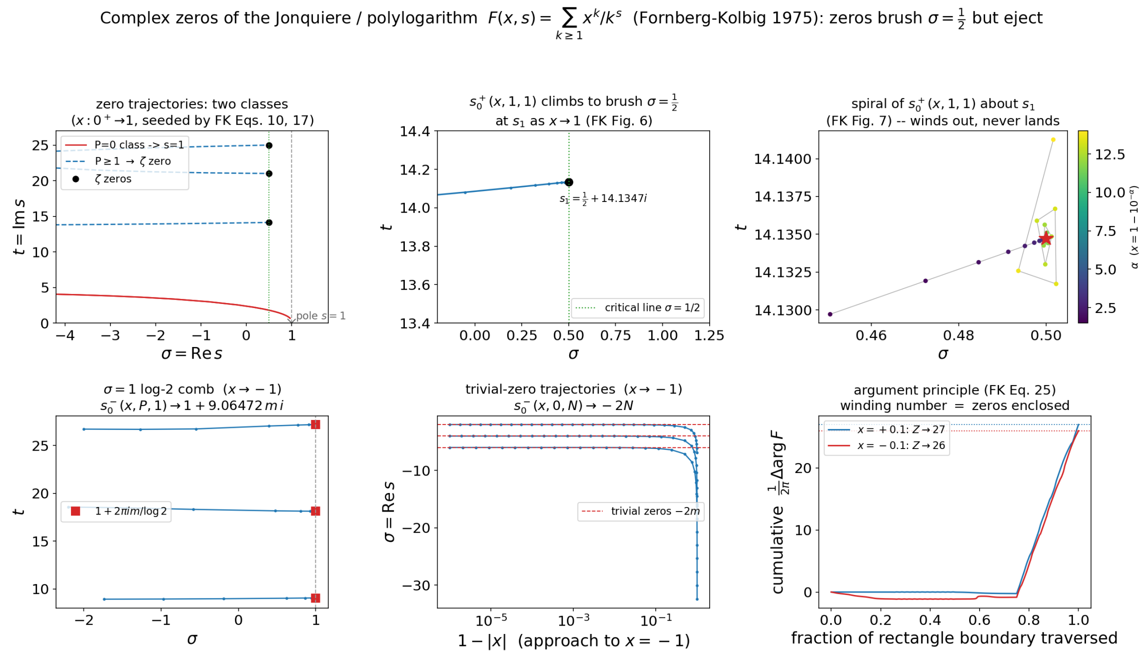

Figure 14: Reproduction of Fornberg and Kölbig [10]: the polylogarithm's zero trajectories as the argument x sweeps $(-1,1)$. The transient $\sigma=\frac{1}{2}$ brush and spiral about s_1 (the money contrast—FK's zeros approach but never land), the $\sigma=1$ log-2 comb at the η endpoint, the trivial-zero trajectories onto $-2m$, and the argument-principle counts. The bridge's knob, by contrast, drives the zeros to *converge* onto $\sigma=\frac{1}{2}$ (Section 7).

not converge onto $\sigma=\frac{1}{2}$: they are born far out in the left half-plane, migrate rightward, and are all eventually absorbed into the pole emerging at $s=1$—some bypassing ζ's zeros, others approaching to within $|s-s_1|\approx 9\times 10^{-7}$ before being *ejected* and spiralling back out to $|s-s_1|\approx 7\times 10^{-3}$, also ending at $s=1$. The mechanism is that the $x\to 1$ expansion (FK Eq. 26) carries a branch term $\Gamma(1-s)(-\ln x)^{s-1}$ whose phase winds as $\ln(-\ln x)\to-\infty$, while $F(x,s)$ is *entire* for $|x|<1$, so the $s=1$ pole only emerges at the endpoint and swallows the migrating zeros. The bridge's harmonic-count knob, off a genuinely zeroless endpoint, is the knob on which this chased convergence actually happens: the zeros land on $\sigma=\frac{1}{2}$ and stay, at $O(1/K)$ (Section 7).

Remark 3. At the *other* endpoint, $x\to -1$, some trajectories *do* coincide with ζ's non-trivial zeros—but only because $F(-1,s)=-\eta(s)$ already carries them. That is the endpoint *having* the zeros, not a dynamical convergence onto the line; the genuine dynamical approach, $x\to +1$, is the transient/spiral one.

A.2 A primitive Dirichlet L-function: $L(s,\chi_4)$

The sharpest test of the bridge is to run its *own* knob—the harmonic truncation K and the warp—on a family with its own critical line and functional equation. We take the odd non-principal character mod 4,

$$L(s,\chi_4)=\sum_{n\geq 1}\chi_4(n)\,n^{-s}=1-3^{-s}+5^{-s}-7^{-s}+\cdots,\qquad \chi_4(n)=\sin(\tfrac{\pi n}{2}),\tag{22}$$

a *primitive* L-function: entire (no pole), root number $W=1$, its own critical line $\mathrm{Re}\,s=\frac{1}{2}$, and the Lerch specialisation $L(\frac{1}{2},\frac{1}{2},s)=2^s L(s,\chi_4)$. The driver `lfunction_bridge.py` runs the bridge on it (Fig. 15); the findings sharpen the taxonomy.

The endpoint is *structured*, not zeroless. The identity $\chi_4(n) = \sin(\pi n/2)$ makes χ_4 a mean-zero periodic coefficient, exactly like η's $(-1)^{n-1} = \cos(\pi n)$. So the bland continuous endpoint is *not* the zeroless $1/(s-1)$ of the ζ side but the structured oscillatory integral

$$G(s) = \int_1^\infty \sin\!\big(\tfrac{\pi x}{2}\big)\, x^{-s}\,\mathrm{d}x, \tag{23}$$

the χ_4 analogue of the continuous-η integral (3), carrying an off-critical Riemann–von Mangoldt zero string that drifts toward $\sigma = \frac{3}{2}$ (the laws (4)–(6) at frequency $\pi/2$). So χ_4 is *η-type*: its zeros migrate from a ground string, they are not born from nothing. (This is why the "zeroless endpoint" of the introduction is specific to the constant-coefficient ζ side; a mean-zero periodic coefficient gives a structured off-critical string instead.)

One line, no companion—cleaner than η. The additive comb $L^{(K)} = G + \frac{1}{2} - \sum_{k\le K} D_k/(\pi k) \to L(s,\chi_4)$ (the sawtooth modes D_k now living at the odd-*quarter* frequencies $(2k \pm \frac{1}{2})\pi$, where η's sat at $(2k \pm 1)\pi$) births the zeros onto $\sigma = \frac{1}{2}$ at the same $O(1/K)$ rate: the $K = 0$ ground string near $\mathrm{Re} \in [0.80, 0.96]$ reaches $\mathrm{Re} \approx 0.50$ by $K = 45$. But because $L(s,\chi_4)$ is *primitive*—it does not factor as (prefactor) $\cdot\,\zeta$—there is *no companion line*: where η split onto $\sigma = \frac{1}{2} \cup \sigma = 1$ and the ζ-warp onto $\sigma = \frac{1}{2} \cup \sigma = 0$, the whole G-ground string maps 1:1 onto the single line $\sigma = \frac{1}{2}$. The densities match on the nose (the conductor $q = 4$ supplies exactly the doubling η got from its $\sigma = 1$ comb; 5 ground-string zeros = 5 genuine L zeros for $0 < t < 20$), so the migration is a clean bijection with no split to bookkeep.

The warp: a Lerch tie-in, and the exact dual of the η-warp. Warping the phase-$\frac{1}{2}$ carrier $\sin(\pi y)$ so the cells collapse onto the midpoints lands on

$$2^s L(s,\chi_4) = L(\tfrac{1}{2}, \tfrac{1}{2}, s) = \Phi(-1, s, \tfrac{1}{2}), \tag{24}$$

the Lerch specialisation (the Lerch family of 12), with the load-bearing lower-endpoint term $+2^s$ the same half-weight the ζ-warp carried. This is the exact half-cell-shifted *dual* of the η-warp (Section 5.5): there the cos carrier's integer phase $\alpha = 1$ survived and the midpoint $\alpha = \frac{1}{2}$ was annihilated; here the sin carrier's midpoint $\alpha = \frac{1}{2}$ survives and $\alpha = 1$ dies. All zeros lie on the single $\sigma = \frac{1}{2}$ (2^s never vanishes, so again no companion).

Remark 4 (A modulus surprise)**.** Warping the *genuine* χ_4 carrier $\sin(\pi x/2)$ at the midpoints lands not on χ_4 but on a *mod*-8 character: $\sum_m \sin\!\big(\frac{\pi}{2}(m+\frac{1}{2})\big)(m+\frac{1}{2})^{-s} = 2^s \frac{\sqrt{2}}{2}\, L(s,\chi_{-8})$, with χ_{-8} the odd character mod 8 (verified as an identity to machine precision). Sampling a period-q carrier on the half-integer lattice can reveal a period-$2q$ character.

This is the one case in the appendix where the bridge's *own* machinery is run forward on a new family, and it is the load-bearing data point for the top row of Table 2: the born-and-migrate-*onto*-$\frac{1}{2}$ behaviour survives for a genuine primitive Dirichlet L-function, and is in fact *cleaner* there than for η (a single line, no companion). It is a computational demonstration, not a theorem—the migration is exhibited numerically, with GRH assumed for the target's zeros as usual—and it uses the bridge's discreteness knob, not FK's argument knob.

A.3 The published α-sweep analogues: Hurwitz and Lerch

The closest *published* deform-and-track studies vary a continuous *parameter* of a Hurwitz or Lerch zeta rather than a harmonic count. The driver `hurwitz_lerch_zeros.py` reproduces the two nearest ones and unifies them (Fig. 16).

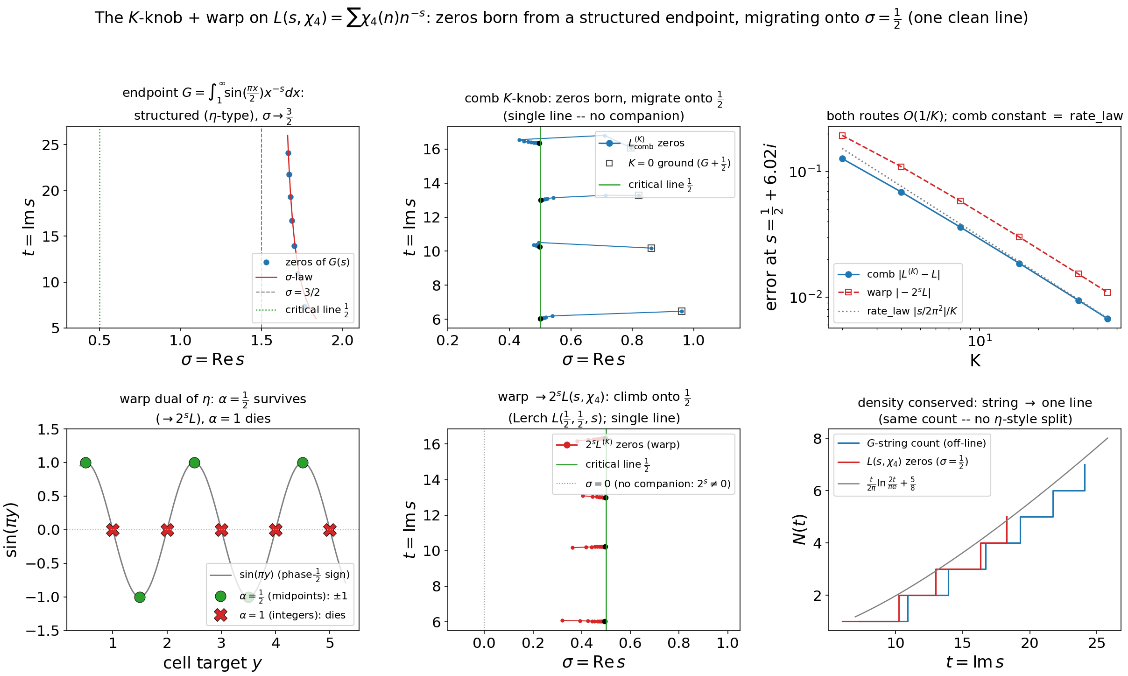

Figure 15: The bridge's own knob on the primitive Dirichlet L-function $L(s,\chi_4)$. The structured endpoint $G(s)$ (23) supplies an off-critical ground string ($\sigma \to \frac{3}{2}$); the K-knob migrates it onto the single line $\sigma = \frac{1}{2}$ at $O(1/K)$, with *no companion* (primitivity), and the phase-$\frac{1}{2}$ warp lands on $2^s L(s,\chi_4) = L(\frac{1}{2},\frac{1}{2},s)$ (24). The born-onto-$\frac{1}{2}$ arrow of the taxonomy, surviving beyond ζ/η.

Garunkštis–Steuding: stable and unstable Hurwitz trajectories. Garunkštis and Steuding [11] track the zeros of the Hurwitz zeta $\zeta(s,\alpha)$ as the shift α sweeps from 1 (where $\zeta(s,1) = \zeta(s)$) to $\frac{1}{2}$ (the half-shift), calling a zero *stable* when its trajectory begins and ends on $\sigma = \frac{1}{2}$. Newton-continuing the first eight Riemann zeros in α reproduces the split: *five* are stable (they bulge off $\frac{1}{2}$—one wanders to Re ≈ 0.9, another to Re < 0—then return to a Riemann zero at $\alpha = \frac{1}{2}$), and *three* are unstable, landing exactly on the $\sigma = 0$ companion comb $s = 2\pi i m/\ln 2 = 9.0647\, m\, i$ (the $2^s - 1$ zeros the half-shift carries, (15)). Extending below $\alpha = \frac{1}{2}$ exhibits the Davenport–Heilbronn off-line wandering—the continuous-trajectory view of the α-scatter in Fig. 8.

The Lerch family square: three ancestors in one picture. The two-parameter Lerch zeta $L(\lambda,\alpha,s) = \sum_{m\geq 0} e^{2\pi i\lambda m}(m+\alpha)^{-s}$ holds the bridge's four objects at the corners of the (λ,α) unit square, all verified to $\sim 10^{-31}$:

$$L(1,1,s) = \zeta(s), \quad L(1,\tfrac{1}{2},s) = (2^s-1)\zeta(s), \quad L(\tfrac{1}{2},1,s) = \eta(s), \quad L(\tfrac{1}{2},\tfrac{1}{2},s) = 2^s L(s,\chi_4). \tag{25}$$

Its three deformation edges are exactly the three published deform-and-track programs, each landing on a *different* companion line:

- the $\lambda = 1$ edge ($\alpha : 1 \to \frac{1}{2}$) is the Garunkštis and Steuding [11] Hurwitz shift, $\to (2^s-1)\zeta$, with its companion on $\sigma = 0$;
- the $\alpha = 1$ edge ($\lambda : 1 \to \frac{1}{2}$) is Fornberg and Kölbig [10] in disguise—$z = e^{2\pi i\lambda}$ sweeps the unit circle from 1 to -1, and $F(-1,s) = -\eta \longrightarrow \eta$, companion on $\sigma = 1$;
- the $\lambda = \alpha$ diagonal ($1 \to \frac{1}{2}$) is the equal-parameter sweep of Garunkštis and Tamošiūnas [12], $\to 2^s L(s,\chi_4)$, with *no* companion: $L(s,\chi_4)$ is primitive, so the first three Riemann zeros ride down onto the first three $L(s,\chi_4)$ zeros $(6.02, 10.24, 12.99)$ without ever leaving $\sigma = \frac{1}{2}$.

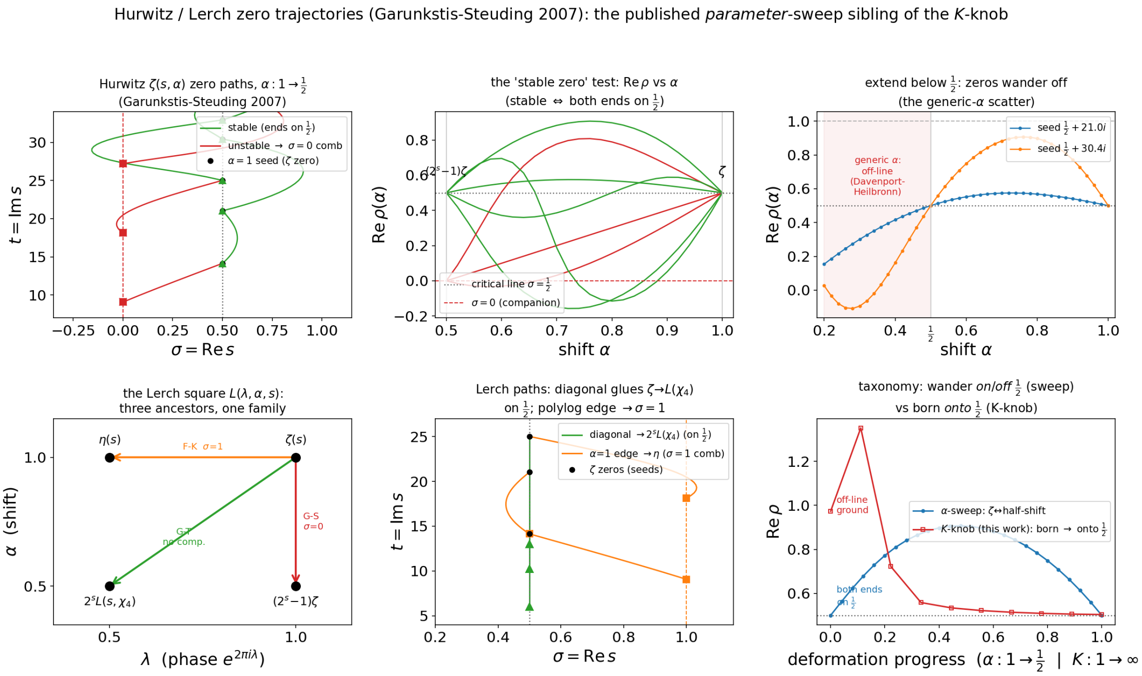

Figure 16: Reproduction of the closest published analogues: Garunkštis and Steuding [11] Hurwitz α-trajectories (stable zeros returning to $\sigma = \frac{1}{2}$; unstable ones landing on the $\sigma = 0$ comb) and the Garunkštis and Tamošiūnas [12] $\lambda = \alpha$ diagonal, inside the Lerch family square (25) whose three edges are the three published ancestors. The last panel contrasts a parameter-sweep's *on/off*-$\frac{1}{2}$ wandering with the K-knob's *onto*-$\frac{1}{2}$ birth.

One family thus knits Fornberg–Kölbig, Garunkštis–Steuding, Garunkštis–Tamošiūnas, and the bridge's ζ, half-shift, η, and χ_4 objects together, with the companion line ($\sigma = 0$, $\sigma = 1$, or none) reading off which edge one took. (An evaluator note: `mpmath`'s `lerchphi` diverges as $z \to 1$; we splice in the Hurwitz zeta there, since $L(\lambda, \alpha, s) \to \zeta(s, \alpha)$ at integer λ.)

The contrast with the K-knob. The α- and λ-sweeps deform between two configurations that *both* already carry infinitely many zeros—no zero is *born*; a "stable" zero merely leaves $\frac{1}{2}$ and returns, its arrow pointing *on and off* the line. The bridge's K-knob starts from a zeroless or structured endpoint, the zeros are *born*, and they climb *onto* $\frac{1}{2}$ and stay ($O(1/K)$). The driver's final panel overlays a stable α-trajectory (both ends on $\frac{1}{2}$, wandering between) against a genuine $\zeta^{(K)}$ trajectory (off-line at $K = 1$, onto $\frac{1}{2}$): the middle row of Table 2 beside the top row, in one axis. Nothing here is claimed as new—G–S and G–T are reproduced, and the family-square unification is expository, a re-presentation of known objects.

A.4 The reverse-direction foil: Epstein lattice zeta

The sharpest single foil is a family in which a continuous knob *generates* off-critical zeros that move the *opposite* way to the bridge's. The driver `epstein_zeros.py` reproduces Travěnec and Šamaj [36] on the hypercubic Epstein zeta

$$Z_d(s) = \sum_{n \in \mathbb{Z}^d}{}' |n|^{-2s}, \tag{26}$$

with the spatial dimension d as a continuous (analytically continued) knob. We evaluate it through the Terras/theta completed form

$$\Lambda_d(s) = \pi^{-s}\Gamma(s)Z_d(s) = \int_1^{\infty}\big[t^{s-1} + t^{d/2-s-1}\big]\big(\psi(t)^d - 1\big)\,\mathrm{d}t + \frac{1}{s-d/2} - \frac{1}{s}, \tag{27}$$

where $\psi(t) = \sum_{n\in\mathbb{Z}} \mathrm{e}^{-\pi n^2 t}$ is a Jacobi theta and $\psi(t)^d$ for *real* d is Travěnec–Šamaj's continuation in the dimension. This is fast, entire except for poles, and valid for all s; it is validated against the direct lattice sum ($\operatorname{Re} s > d/2$), the closed forms $Z_1(s) = 2\zeta(2s)$ and $Z_2(s) = 4\zeta(s)\beta(s)$, the functional equation $\Lambda_d(s) = \Lambda_d(d/2-s)$ (critical line $\sigma = d/4$), and the pole residue $\pi^{d/2}/\Gamma(d/2)$, all to $\sim 10^{-29}$.

A pitchfork at the critical dimension. Because Λ_d is even about the real centre $s = d/4$, that centre is a *double* zero exactly when $g(d) = \Lambda_d(d/4) = 0$; its root is reproduced to ten digits as $d^* = 9.2455524667$ (Travěnec–Šamaj's 9.24555). The sign of $\Lambda_d(d/4)\,\Lambda_d''(d/4)$ decides the branch: for $d < d^*$ the lowest pair sits *on* the critical line at $d/4 \pm \mathrm{i}\,t_1(d)$ with $t_1(d) \to 0$ as $d \uparrow d^*$ (they slide down the line to collide); at d^* they meet at the real centre; and for $d > d^*$ they split into a *real* off-critical pair $d/4 \pm \delta(d)$ that leaves the line and migrates to the strip boundaries $\{0, d/2\}$ (at $d = 25$, $\sigma_- \approx 2 \times 10^{-4}$ and $\sigma_+ \approx d/2 - 2 \times 10^{-4}$). An argument-principle count certifies this independently of any root-finder: *zero* real zeros in the strip for $d < d^*$, exactly *two* for $d > d^*$.

The reverse arrow. This is the bottom row of Table 2 made concrete: zeros *born off* $\sigma = d/4$ and *migrating away* to the strip edges—the opposite of the bridge's zeroless-→-born-→-onto-$\frac{1}{2}$. The driver's final panel overlays the Epstein real zero (born off, moving away as $d : d^* \to 25$) against a genuine $\zeta^{(K)}$ zero (born from the zeroless endpoint, climbing onto $\frac{1}{2}$ as $K : 1 \to 89$) on one axis of $\operatorname{Re} s -$ (critical line): two arrows, opposite ways. The caveat that keeps the contrast honest is that Epstein zeta lacks an Euler product (the sum (26) is geometric, over a lattice, not arithmetic), so this is a *cross-family* foil about the *breadth* of the directional phenomenon, not a shared arithmetic mechanism. A two-dimensional shape-knob variant moves the zeros the same way under a lattice deformation [4]. As with the other three cases, this is a faithful reproduction plus the contrast figure, not a claim: the generation of off-critical zeros is Travěnec–Šamaj's result.

What the four cases establish. Together they locate the bridge inside its lineage without over-claiming. The born-onto-$\frac{1}{2}$-from-a-zeroless-start arrow (top row of Table 2) is unmatched in the deform-and-track literature for any family other than ζ and η: its two published relatives (Appendices A.1 and A.3) merely *wander* on and off the line between two zero-rich ends, and the closest lattice-zeta analogue (Appendix A.4) drives its zeros the opposite way. That the same arrow *does* survive for a genuine primitive Dirichlet L-function (Appendix A.2) is the appendix's one forward step beyond reproduction—and even there the claim is a computational demonstration, not a theorem.

B The geometric series: the bridge in miniature

The four cases above are *lateral* moves—other Dirichlet-type objects with their own zeros. This last appendix is a *vertical* one: the same construction run on the simplest possible summand, the geometric series, where every step collapses to an *elementary closed form* and there are **no special functions and no zeros** anywhere. Nothing here is new mathematics—the one identity that does the work is the classical coth partial fraction, and the summation of divergent series to their analytic values is textbook Abel/Euler summability [cf. 17]. Its purpose is a

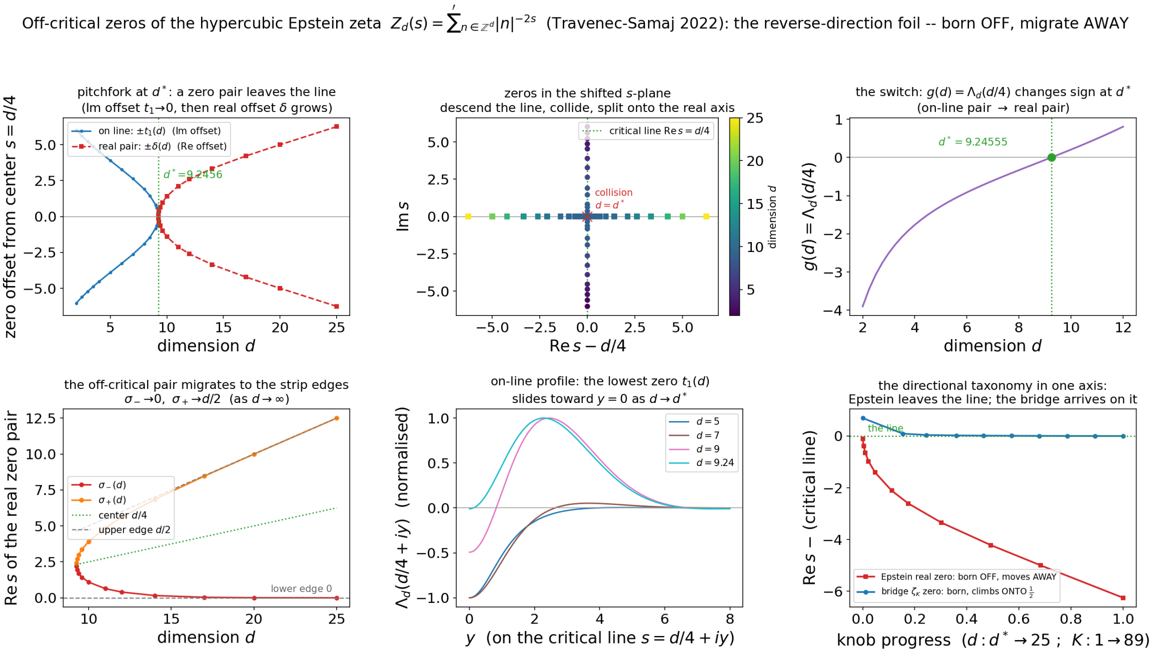

Figure 17: Reproduction of the reverse-direction foil [36]. The hypercubic Epstein zeta Z_d (26), evaluated through the completed form (27), undergoes a pitchfork at $d^* = 9.24555\ldots$: for $d > d^*$ a real pair is born *off* $\sigma = d/4$ and migrates *away* to the strip edges. The last panel sets this reverse arrow beside the bridge's born-*onto*-$\frac{1}{2}$ motion.

fully transparent *consistency check* of the apparatus, and it renders the arc's subtlest point—convergence *beyond* the classical domain (Section 7.1)—completely elementary.

The two endpoints. Take the ratio s as the variable and $f(x) = s^x = e^{x\ln s}$. The discrete object and its continuous shadow are both elementary,

$$g(s) = \sum_{n\geq 0} s^n = \frac{1}{1-s}, \qquad C(s) = \int_0^\infty s^x\,dx = -\frac{1}{\ln s} \qquad (|s| < 1,\ \text{else by continuation}), \tag{28}$$

and both keep the pole at $s = 1$ with residue -1: near $s = 1$, $-1/\ln s = -1/(s-1) - \frac{1}{2} + O(s-1)$, the bland pole of Section 2 in a new guise.

The comb telescopes. Euler–Maclaurin with the periodic-Bernoulli sawtooth $\{x\} - \frac{1}{2} = -\sum_{k\geq 1}\sin(2\pi kx)/(\pi k)$ and $\int_0^\infty \sin(2\pi kx)\,e^{x\ln s}\,dx = 2\pi k/((\ln s)^2 + 4\pi^2k^2)$ gives, term by term, the interpolant

$$g_K(s) = -\frac{1}{\ln s} + \frac{1}{2} - \sum_{k=1}^{K}\frac{2\ln s}{(\ln s)^2 + 4\pi^2k^2}, \tag{29}$$

whose $K = 0$ anchor is the shadow plus the load-bearing half-weight $\frac{1}{2}$—the same term that cancels the shadow's $-\frac{1}{2}$ at the pole, so *every* g_K inherits the residue -1. The full comb **telescopes exactly**, by the coth partial-fraction expansion $\sum_{k\geq 1} 2L/(L^2 + 4\pi^2k^2) = \frac{1}{2}\coth\frac{L}{2} - \frac{1}{L}$ (equivalently the generating function of the Bernoulli numbers; 9), giving

$$g_\infty(s) = \tfrac{1}{2} - \tfrac{1}{2}\coth\tfrac{\ln s}{2} = \frac{1}{1-s}, \tag{30}$$

the discrete sum, on the nose.

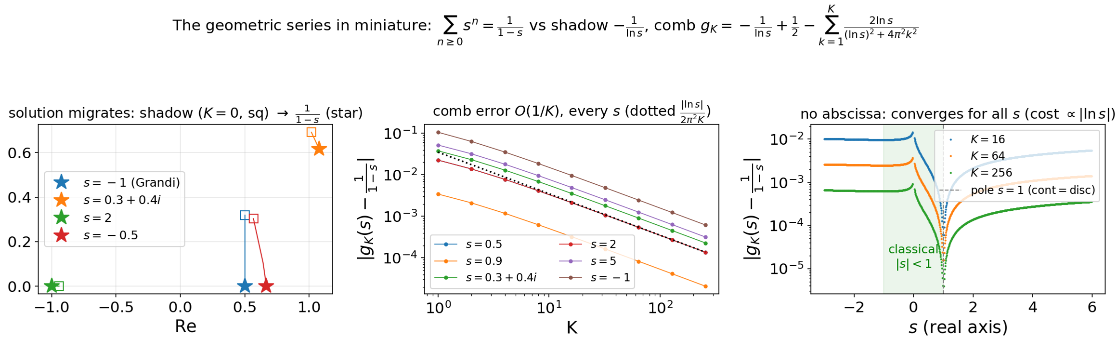

Figure 18: The geometric series in miniature (`geometric_bridge.py`). *Left:* with no zeros to move, the *value* $g_K(s)$ migrates from the continuous shadow ($K = 0$, squares) onto the discrete sum $1/(1-s)$ (stars); at $s = -1$ it slides down $\mathrm{Re} = \frac{1}{2}$ from i/π to Grandi's $\frac{1}{2}$. *Centre:* the comb error is $O(1/K)$ for every s, tracking $|\ln s|/(2\pi^2 K)$ (31)—the arc's rate law with $s \mapsto \ln s$. *Right:* no abscissa—the error is smooth *through* the classical boundary $|s| = 1$, dipping to 0 at the pole (shadow = sum) and growing like $|\ln s|$; for $|s| > 1$ the interpolant sums the divergent series to $1/(1-s)$.

The rate law, with $s \mapsto \ln s$; and no abscissa. The k-th harmonic is $2\ln s/((\ln s)^2 + 4\pi^2 k^2) \sim (\ln s)/(2\pi^2 k^2)$: the same $1/k^2$ sawtooth tail as the ζ-comb term $\sim s/(2\pi^2 k^2)$ of Section 7, with the per-mode constant s replaced by $\ln s$. Hence the identical $O(1/K)$ error law,

$$\left|g_K(s) - \tfrac{1}{1-s}\right| = \frac{|\ln s|}{2\pi^2}\Big(\frac{1}{K} - \frac{1}{2K^2} + \cdots\Big), \tag{31}$$

with the "height/cost" coordinate now $|\ln s|$—the log-distance of s from the pole—in place of $|s| \sim t$. And, exactly as in Section 7.1, each g_K is meromorphic on all of $\mathbb{C}$ with a single pole at $s = 1$: there is *no abscissa of convergence*. So (29) converges for $|s| > 1$ as well, summing the *divergent* series to its analytic value—$1 + 2 + 4 + \cdots = -1$, Grandi's $1 - 1 + 1 - \cdots = \frac{1}{2}$ (Fig. 18, right)—the elementary face of the "$\zeta^{(K)}$ converges left of every abscissa" phenomenon.

The warp factorizes; and runs both ways. The nonlinear route of Section 5 also collapses. Warping $x \mapsto x + \phi_K(x)$ with $\phi_K(x) = \sum_{k=1}^{K} \sin(2\pi k x)/(\pi k)$ and the integer-phase offset $-\frac{1}{2}$ pins each cell onto its integer; because $s^{m+r} = s^m s^r$, the cell index *factors out* of the sum,

$$\int_0^\infty s^{x-\frac{1}{2}+\phi_K(x)}\,dx = \frac{1}{1-s}\,J_K(s), \qquad J_K(s) = \int_0^1 s^{u-\frac{1}{2}+\phi_K(u)}\,du \;\longrightarrow\; 1, \tag{32}$$

so the entire warp convergence is the single scalar $J_K \to 1$—no Hurwitz zeta, no moment machinery (contrast Section 5). The midpoint-phase warp lands instead on $s^{1/2}/(1-s) = \sum_{m\geq 0} s^{m+1/2}$, the elementary echo of the half-shift $\zeta(s, \frac{1}{2})$. Finally, the same harmonic family runs *backwards*: added rather than subtracted, it carries the discrete sum back to the shadow $(\frac{1}{1-s} - \frac{1}{2} + \sum_{k\leq K} h_k \to -1/\ln s)$, so continuous ↔ discrete is one comb read in either direction.

What the miniature establishes. Precisely because it has *no* zeros, functional equation, or arithmetic, it isolates what the comb and warp actually are: a reshaping of the discrete ↔ continuous *measure*, independent of any zero story. Pole preservation, the $O(1/K)$ rate, the no-abscissa stability, and the comb/warp duality all survive the strip-down to elementary functions; the zeros of the main arc are what this scaffold *carries*, not what it *is*. It is a consistency check and an expository on-ramp, not a result.

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